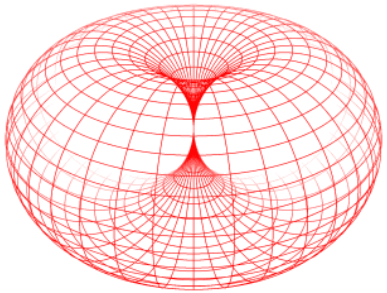
HoT Identities and Vicious Abstractions

in the (Homotopic) Foundations of Mathematics

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« Form is emptiness (*śūnyatā*), emptiness is form. »

The Heart Sūtra

What is Mathematics?

- Ontic answer (« Platonic realism »)
- Ontologic answer (e.g. Badiou)
- Transcendental answer (e.g. Frege).

More standard classification:

.Platonic realism (Gödel),
.Logicism (Frege, Russell),
.Intuitionism (Brouwer, Heyting),
.Formalism (Hilbert)
.Structuralism.
.etc.

1. Mathematical « Platonism »

Mathematics is a science that studies a particular realm of entities among others - that of abstract, *ideal* or *supersensible objects* -, which exist in themselves in the sense that they are independent of the cognitive activity of the mathematicians (e.g. Gödel).

« [...] the objects and theorems of mathematics are as objective and independent of our free choice and our creative acts as is the physical world. [...] the Platonistic view is the only one tenable. Thereby I mean the view that mathematics describes a non-sensual reality, which exists independently both of the acts and [of] the dispositions of the human mind and is only perceived, and probably perceived very incompletely, by the human mind. »

K. Gödel, *Some basic theorems on the foundations of mathematics and their implications*.

« Platonists will be those who consider mathematics to be the *discovery* of truths about structures that exist independently of the activity or thought of mathematicians. »

P. Benaceraff and H. Putnam, Introduction to *Philosophy of Mathematics*, p. 18.

Against the « vulgar » understanding of Platonism

« In the open debate between formalist and intuitionist, since the discovery of the transfinite, mathematicians have become accustomed to summarily designate under the name Platonism any philosophy for which the existence of a mathematical entity is taken as assured, even though this entity could not be built in a finite number of steps. It goes without saying that this is a superficial knowledge of Platonism, and that we do not believe ourselves to be referring to it. All modern Plato commentators on the contrary insists on the fact that Ideas are not immobile and irreducible essences of an intelligible world, but that they are related to each other according to the schemas of a superior dialectic that presides over their arrival. »

A. Lautman, *Essay on the Notions of Structure and Existence in Mathematics.*

Objection to mathematical « Platonism »

This stance does not explain

« The Unreasonable Effectiveness of Mathematics

in the Natural Sciences » (E. Wigner).

That is, it does not explain its transversality

with respect to different domains of entities

(e.g. physics, biology, economics).

For Kant,

mathematics is not so much a science

as the quintessence of scientificity as such.

« […] in any special doctrine of nature there can be only as much *proper* science as there is *mathematics* therein. »

(E. Kant, *Metaphysical Foundations of Natural Science*)

2. Mathematics is Ontology

« Mathematics *is* ontology, the science of being qua being. »

A. Badiou, *Being and Event*, p.4.

« […] in terms of the 'application' of mathematics to the so-called natural sciences […] the clarification is immediately evident if mathematics is the science, in any case, of everything that is, *insofar as it is*. »

A. Badiou, *Being and Event*, p.7.

For instance, the thesis

« No entity without identity » (Quine)

is an ontological claim about what it means to be an entity.

Other ontological categories:

- Multiplicity (being-many).
- Relation (being-with).
- Extension (being-there).
- Quality (being-thus).
- Quantity (being-thus-much).
- Difference (being-different).
- Continuity (being-unbreakable).

3. Transcendental Answer

Mathematics is an activity by means of which

we can refine and enhance

the rational faculties and categories

of human understanding in order to perform specialized tasks.

According to Frege's analogies,

logic is to human language

what a microscope is to the eye

or what a robotic hand is to the human hand.

« I believe that I can best make the relation between my ideography to ordinary language clear if I compare it to that which the microscope has to the eye. Because the range of its possible uses and the versatility with which it can adapt to the most diverse circumstances, the eye is far superior to the microscope. Considered as an optical instrument, to be sure, it exhibits many imperfections, which ordinarily remain unnoticed only on account of its intimate connection with our mental life. But, as soon as scientific goals demand great sharpness of resolution, the eye proves to be insufficient. The microscope, on the other hand, is perfectly suited to precisely such goals, but that is just why it is useless for all others. »

G. Frege, *Begriffsschrift*, Preface.

« Language [...] proves to be deficient in the matter of protecting our thinking from error; it does not even fulfill the first demand made on it in this respect, that of being unequivocal. [...] Language is not in such a way dominated by logical laws that compliance with grammar would of itself guarantee the correctness of thought processes. [...] The shortcomings here stressed are caused by a certain softness and instability of language which, on the other hand, constitute the reason for its many-sided usefulness and potentiality for development. In this respect language can be compared to the hand which, despite its adaptability to the most diverse tasks, is still inadequate. We build ourselves artificial hands-tools for special purposes which function more exactly than the hand is capable of doing. And how is this exactness possible? Through the very rigidity and inflexibility of the parts, the lack of which makes the hand so dexterous. The word language is in the same way also not adequate. We have need of a system of symbols from which every ambiguity has been banned and whose rigorous logical form cannot be escaped by the content. »

G. Frege, *On the Scientific Justification of a Concept-Script*.

Rational thinking is not only about logical deductions, but also about:

- collecting in classes or sets,
- ordering,
- counting,
- measuring,
- comparing, i.e. establishing similarities and differences.
- classifying,
- abstracting,
- establishing proportions,
- calculating,
- predicting,
- reidentifying things through change,
- extracting invariants,
- recognizing patterns,
- acting on things (transforming, operating, deforming),
- schematizing, visualizing, diagrammatising.

We can understand the different branches of mathematics

as conceptual and formal technologies

by means of which we can enhance

the faculties of rational cognition

(deducing, computing, counting, operating, abstracting, classifying, predicting, measuring, schematizing, etc.)

« through [an aprioristic] construction of concepts »
(Kant)

In particular, we shall be concerned

with the notions of *identity* and *difference*,

which can be understood either *ontologically*

(« no entity without identity » or principle of identity $a = a$)

or *transcendentally*

(we cannot think without making abstraction of irrelevant factors,
i.e. without 'forgetting' certain differences
and equalizing the corresponding objects).

HoTT can be understood as a far-reaching reconceptualization

of the notions of *identity* and *difference,*

which proceeds by rethinking one of the most fundamental mathematical notions,

namely the notion of equality ‘=’,

which is transversal to all mathematics disciplines (e.g. geometry, algebra, analysis, etc.)

Foundations of Mathematics

What is a mathematical object?

- **Substantialist answer (set theory, 1870s-1920s)**:

A mathematical object is defined by its ***content,***

namely as a *set equipped with a structure*.

- **Structuralist answer (category theory, 1940s)**:

A mathematical object is defined by its relational ***context***,

namely as a *node in a network of relations*.

- **Homotopy Type Theory** aka HoTT **(2000s)** is a sort of synthesis between these two approaches.

The transition between these paradigms is marked

by a progressive reconceptualization of the mathematical notions

of *equality* $a = b$ and *identity* $a = a$.

Equality in Set theory

Equality '=' is a relation that a set can entertain only with itself, that is, equality propositions $a = b$ can always be reduced to identities.

This notion of equality is called *strict equality.*

Equality statements of the form $a = b$ can only be interpreted:

- either as a relation of *synonymy*, where the two names a and b refer to the same set,

- or as an *abuse of notation*.

Equality as synonymy according to Frege

« Identity of content differs from conditionality and negation in that it applies to names and not to contents. Whereas in other contexts signs are merely representatives of their content so that every combination into which they enter expresses only a relation between their respective contents, they suddenly display their own selves when they are combined by means of the sign for identity of content; for it expresses the circumstance that the two names have the same content. Hence, the introduction of a sign for identity of content necessarily produces a bifurcation in the meaning of all signs: they stand at times for their content, at times for themselves. »

G. Frege, *Begriffsschrift*, 1879, §8.

Equality as synonymy according to Quine

« Identity is such a simple and fundamental idea that it is hard to explain otherwise than through mere synonyms. To say that *x* and *y* are identical is to say that they are the same thing. Everything is identical with itself and with nothing else. But despite its simplicity, identity invites confusion. E.g., it may be asked: Of what use is the notion of identity if identifying an object with itself is trivial and identifying it with anything else is false? [...] For truth of a statement of identity it is necessary only that '=' appear between names of the same object; the names may, and in useful cases will, themselves be different. For it is not the names that are affirmed to be identical, it is the things named. [...] Still, since the useful statements of identity are those in which the named objects are the same and the names are different, it is only because of a peculiarity of language that the notion of identity is needed. If our language were so perfect a copy of its subject matter that each thing had but one name, then statements of identity would indeed be useless.[...] The utility of language lies partly in its very failure to copy reality in any one-thing-one-name fashion. The notion of identity is then needed to take up the slack. [...] what are identical are the objects with themselves and not the names with one another; the names stand in the statement of identity, but it is the named objects that are identified.[...] Statements of identity consisting of '=' flanked by singular terms are needed, we saw, because language includes a redundancy of names. »

W.V.O. Quine, *Methods of Logic*, § 35.

« [...] the statements of identity that are true and not idle consist of unlike singular terms that refer to the same thing. »

W.V.O. Quine, *Word and Object*, p. 107.

Equality as an « abuse of notation »

« This equality $2 \times 3 = 5 + 1$ makes sense in the mainstream of mathematics by saying that the two sides denote the same integer [...]. This is […] undoubtedly correct, but it misses the essential point: There is a finite *computation* process which shows that the denotations are equal. It is an abuse [...] to say that 2×3 *equals* $5 + 1$ […] we must *do* something (make a proof or a calculation [...]) to show that these two [expressions] have the same *denotation*. »

J.-Y Girard, *Proofs and Types*, 1990.

Stretching equalities in category theory and HoTT

Strict equality '=' is « stretched » to a new relation

(called *isomorphism* in category theory
and *propositional equality* in HoTT)

that can relate different objects.

The main differences between strict and stretched equalities are that:

- Stretched equalities can relate numerically different entities *a* and *b*.
- There might be many different stretched equalities between *a* and *b*.

Set-Theoretic Foundations

Mathematical Object
=
Set + Structure + Laws

Every mathematical objects is a kind of *structured dust*.

Morphism between such structured sets

=

function between the underlying sets that respects the structure.

« [...] the decisive break—in which mathematics blindly pronounces on its own essence—is Cantor's creation. It is there alone that it is finally declared that, despite the prodigious variety of mathematical ‘objects’ and ‘structures’, they can all be designated as pure multiplicities built, in a regulated manner, on the basis of the void-set alone. »

A. Badiou, *Being and Event*, p. 6.

Algebraic Example of a Structured Set: Groups (E. Galois, XIX century)

A group $(X, *, e)$ is

- a set X (***data***)
- enriched with a *closed binary operation* $*$ and a *distinguished element* e (***structure***);
- such that (***laws***)

.the distinguished element e is the neutral element for the operation $*$:

$$g * e = e * g = g$$

.for each $g \in X$, there is an element $g^{-1} \in X$ (called the inverse of g) such that

$$g^{-1} * g = g * g^{-1} = e.$$

. the operation $*$ is associative:

$$(g * h) * k = g * (h * k).$$

A morphism of groups $f : (G, *, e_G) \to (H, \star, e_H)$ is a function btw the underlying sets $G \to H$ that respects the groups structure, i.e. such that

$$f(g * g') = f(g) \star f(g')$$

Example: the entire numbers $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$

$(\mathbb{Z}, +, 0)$ is a group.

- Neutrality of 0: $n + 0 = 0 + n = n$.
- Existence of inverses: for each $n \in \mathbb{Z}$, there is an inverse $-n$ such that $n + (-n) = 0$.
- Associativity: $(n + m) + k = n + (m + k)$.

Counterexample: the natural numbers $\mathbb{N} = \{0, 1, 2, \ldots\}$

- Neutrality of 0 holds.
- Associativity of + holds.
- But there are no inverses !

$\mathbb{N}$ is not a group, but a ***monoid***.

Geometric Example of a Structured Set: Topology (F. Hausdorff, 1914)

• A set X is not a space (e.g. there is no notion of distance between any two elements $x, y \in X$).

• A *topology* is a first kind of structure that we can define on a set to endow it with a (*sub-metric* or *distance-independent*) geometric nature.

• The fundamental topological notion is the notion of ***continuity***,…

> … that is, changes that do not engage *abrupt breaks, jumps,* or *sharp discontinuities*.

• To do so, one understands each element $x \in X$ as a geometric point and defines which are the ***unbreakable neighborhoods*** of x.

- A ***topology*** on a set X is a structure that assigns to each « point » $x \in X$…

  .a collection $\mathcal{N}(x)$ of subsets U of X

  .that will play the role of the unbreakable *neighborhoods of x* (and which satisfy certain axioms).

- A set X endow with a topology $\mathcal{N}$ defines a ***topological space*** $(X, \mathcal{N})$.

- Morphisms between topological spaces are given by ***continuous functions***, which are:

  .maps between the underlying sets

  .that respects the additional structure, i.e. that does not break the unbreakable parts.

Examples of Topological Spaces

- **Discrete topology.**

A *discrete space* (a space with no continuity at all) is a set endowed with the so-called *discrete topology*, in which there is a neighborhood $\{x\}$ for each x

All the points are isolated from each other.

Here there is no community !

- **Indiscrete topology**.

In the indiscrete topology, there is only one neighborhood for each x, namely the whole space X.

The most intimate neighborhood of a point x is the whole set X itself.

Here there is no privacy !

Category Theory

Category Theory Foundations

CT provides a *structuralist*, *relational*, *behavioral*, or s*ociological*

characterization of mathematical objects

that faithfully implements the *Structuralist Credo*:

The *relations* determine the *relata*

Objects are not determined by their content (as in set theory),

but rather by their relational context.

« […] the attitude gradually emerged that the crucial characteristic of mathematical structures is not their internal constitution as set-theoretical entities but rather the relationship among them as embodied in the network of morphisms […] »

J.L. Bell, *Category Theory and the Foundations of Mathematics,* 1981.

Definition of a Category

A *category* $\mathscr{C}$ is given by:

- a collection of *objects* $a, b, c \ldots$
- a collection of arrows or *morphisms* between the objects $a \to b$ such that:

  .two arrows

  $$f : a \to b \quad \text{and} \quad g : b \to c \quad \text{satisfying} \quad target(f) = source(g)$$

  can be composed into a single arrow denoted

  $$g \circ f : a \to c$$

  in an associative manner:

  $$(h \circ g) \circ f = h \circ (g \circ f)$$

  .such that each a is equipped with an identity arrow id_a which acts as a neutral element for the composition:

  $$f \circ id_a = f \quad \text{and} \quad id_b \circ f = f \quad \text{with} \quad f : a \to b.$$

What does it mean that an object is determined by the relations?

- Empiricist definition:

Object = *bundle of properties*.

- ***Principle of Indiscernibility of Equals***:

If two objects are equal they should have the same properties.

- This means that properties have to be invariant across equalities:

$$P(a) \text{ and } a = b \Rightarrow P(b)$$

The Category-Theoretical Stretching of Equality

- In category theory, *strict equalities*

$$a = b$$

(meaning that a and b denote one and the same thing)

are ***stretched*** to *isomorphisms* (« *equal form* »)

$$a \simeq b$$

(meaning that the numerically different objects are considered « equal »
in the corresponding categorical context).

In category theory,

the ***Principle of Indiscernibility of Equals***

becomes

the ***Principle of Indiscernibility of Isomorphic Objects***:

Isomorphic objects have the same intrinsic properties.

or

An intrinsic property of an object in a category

is a property that is invariant under isomorphisms.

The « relations determine the objects » in the precise sense that what an object is

— i.e. the intrinsic properties it has —

depends on the relations that it entertains with other objects.

On Isomorphisms and the Indiscernibility of Identicals

« [...] two isomorphic structures are, essentially, the same. [...] (Principle of Isomorphism) all grammatically correct properties of objects of a fixed category are to be invariant under isomorphism. Here, a property $P(Y)$ of the variable object Y in a given category is invariant under isomorphism if $P(Y)$ and $Z = Y$ imply $P(Z)$.The just-stated principle is the analog, formulated for isomorphism in place of ordinary equality, of Leibniz's indiscernibility of identicals, the usual axiom of equality that allows to substitute any thing by any other thing equal to it. [...] The basic character of the Principle of Isomorphism is that of a constraint on the language of Abstract Mathematics; a welcome one, since it provides for the separation of sense from nonsense. »

M. Makkai, *Towards a Categorical Foundation of Mathematics*.

Structuralism according to Hegel

« These many diverse things stand in essential reciprocal action by virtue of their properties; the property is this reciprocal connecting reference itself, apart from which the thing is nothing. »

G.W.F. Hegel, *The Science of Logic*.

Isomorphisms

• An *isomorphism* is a morphism

$$f : a \to b$$

such that there exists another morphism

$$f^{-1} : b \to a$$

satisfying

$$f^{-1} \circ f = id_a \qquad \text{and} \qquad f \circ f^{-1} = id_b$$

• An isomorphism is a kind of internal decomposition (or factorization) of the trivial self-identity id_a that passes through an external object b.

Examples of Isomorphisms

• **Isomorphisms** $f : X \to Y$ **in the category of sets**:

Bijective (or one-to-one) functions.

• **Isomorphisms** $f : (G, *, e_G) \to (H, \star, e_H)$ **in the category of groups** :

Bijective functions btw the underlying sets $G \to H$ that respects the groups structure, i.e. such that

$$f(g * g') = f(g) \star f(g')$$

• **Isomorphisms** $f : (X, \tau) \to (Y, \eta)$ **in the category of topological spaces (called *homeomorphisms*)** :

Bijective functions btw the underlying sets $f : X \to Y$ that respect the topological structure, i.e. such that f and f^{-1} are continuous.

Canonical joke: a topologist is a person that cannot make the difference between a *cup* and a *doughnut* (since one can be bijectively and continuously deformed into the other one).

The Conceptual Kernel of Category Theory

- The ***Stretching Equality***:



- ***Principle of the Indiscernibility of Isomorphic Objects***:

The intrinsic properties of an object

have to be invariant under isomorphisms.

Higher Category Theory

The Pitfalls of Category Theory

• Category theory stretches the notion of strict equality $=$ ***between objects*** to the notion of isomorphism $\simeq$.

• However, strict equalities are still used in the very definition of a category in the form of strict equalities ***between morphisms***:

.the associativity of composition

$$(h \circ g) \circ f = h \circ (g \circ f)$$

.the neutrality of the trivial identities wrt compositions:

$$f \circ id_a = f$$

• Moreover, the definition of isomorphisms also uses strict equalities between morphisms:

$$f^{-1} \circ f = id_a$$

Category theory does not implement

a full-fledged stretching of strict equalities into isomorphisms.



Category theory remains quite strict after all.

Stretching all the way up

• In order to solve this problem, we should stretch the strict equalities in the definition of a category:

$$(h \circ g) \circ f = h \circ (f \circ g)$$

$$f \circ id_a = f$$

and in the definition of an isomorphism:

$$f^{-1} \circ f = id_a$$

into isomorphisms $\simeq$.

• Now, the entities related by such equalities (e.g. $f^{-1} \circ f$ and id_a) are already morphisms.

• So the new isomorphisms should be isomorphisms between morphisms !!!

A full-fledged or maximally stretched replacement of equalities with isomorphisms

can only be performed if we make the leap of faith towards ***higher categories***,

i.e. towards a theory of ***categories with higher morphisms***,

i.e., with morphisms between morphisms between morphisms,…

Higher Category Theory

An n-category $\mathscr{C}$ is given by:

.a collection of *objects* $a, b, c \ldots$

.a collection of 1-morphisms $f : a \to b$ between objects.

.a collection of 2-morphisms $H : f \Rightarrow g$ between 1-morphisms f and g with the same source a and target b.

.a collection of 3-morphisms $\eta : H \Rrightarrow K$ between 2-morphisms H and K between 1-morphisms f and g with the same source a and target b.

- And so on and so forth all the way up to n-morphisms.

- In particular:

.0-categories: sets.

.1-categories: ordinary categories.

Higher compositions

- In higher category theory, n-morphisms can meet along entities of any dimension from 0 to $n-1$.

- Hence, there are $p = 0,...,n-1$ manners to compose n-morphisms.

- For instance, 2-morphisms can meet at objects and at 1-morphisms.

- It follows that 2-morphisms can be composed

  .along objects (called *horizontal composition*)
  .along 1-morphisms (called *vertical composition*).

2-categories
0-composition




2-categories
1-composition


3-categories
0-composition


3-categories 1-composition


3-categories 2-composition

From globular sets to magmas

- ***Globular set***:

  .Collection of 0-cells (objects), 1-cells (morphisms), 2-cells (2-morphisms), …, n-cells, …

  .Such that each each n-cell is equipped with a $(n+1)$-putative identity cell.

  .Satisfying certain « globularity conditions ».

- We enrich the globular set by adding all possible binary compositions.

- This yields a lawless or incoherent ***magma***.

Compositions of multi-composable diagrams

- The composition of certain « ambiguous » diagrams can be performed in different orders.

- 1-categorical example:

.In the « ambiguous » diagram,

$$a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$$

there are two possible manners to perform the composition, namely

$$h \circ (g \circ f) \qquad \text{and} \qquad (h \circ g) \circ f.$$

- 2-categorical example: interchange law.

Stretching or Straightening

- **Straightening (or juridical) option:**

We can impose ***coherence laws*** according to which

the different possibles compositions of a multi-composable diagram

should be strictly equal.

This yields ***Strict Higher Categories***

- **Stretching (or anarchist) option**:

The different possibles compositions

of a multi-composable diagram

should be related by morphisms one level higher (***coherators***).

This yields ***Weak Higher Categories***

Examples of Coherators

- Associators:

$$h \circ (g \circ f) = (h \circ g) \circ f \rightsquigarrow assoc_{h,g,f} : h \circ (g \circ f) \longrightarrow (h \circ g) \circ f)$$

- Unitors:

$$f \circ id_a = f \rightsquigarrow unitor_f : f \circ id_a \longrightarrow f$$

Category-Theoretical Anarchist Credo

We can stretch *strict laws* by adding higher *coherators*.

(e.g. associators, unitors, pentagonators, federators,

mutualisers, mediators, kidtors, cancelators, etc.)

Let's replace transcendent laws with immanent coherators!

Stretching Isomorphisms Further and Further

The correct notion of « sameness » in a higher category

can be obtained by maximally stretching

the 1-categorical notion of isomorphism.

Two objects a and b in a higher category are « equal »

if there is a morphism $f : a \to b$

that is *invertible in the weakest possible sense* of the term.

Towards Maximally Stretched Equalities

- In 0-categories (sets), structs equalities are the *maximally stretched notion of sameness* available.
- In 1-categories, isomorphisms are the *maximally stretched notion of sameness* available.
- Indeed, the definition of isos

$$f^{-1} \circ f = id_a$$

uses a strict equality $=$ between $f^{-1} \circ f$ and id_a, but we do not have 2-isos to stretch further these strict equalities is something like

$$f^{-1} \circ f \simeq id_a$$

- But in higher categories we do have higher isomorphisms to stretch further.

The higher the category,

the more higher « ligaments » we have

to stretch equalities and identities

more and more.

Maximally Stretched Equalities in 2-Categories

- In a 2-category, we now have a notion of 2-isos to stretch the strict equalities

$$f^{-1} \circ f = id_a$$

used to define 1-isos.

- Hence, a maximally stretched equality between two objects in a 2-groupoid is given by a 1-morphism

$$f : x \to y$$

such that there exists a 1-morphism

$$f^{-1} : y \to x$$

such that there exists a 2-iso

$$\alpha : f^{-1} \circ f \to id_x$$

with inverse

$$\alpha^{-1} : id_x \to f^{-1} \circ f$$

such that

$$\alpha^{-1} \circ \alpha = id_{f^{-1} \circ f}$$

Note: these last strict equalities between 2-morphism cannot be stretched since in order to do that we would need 3 -morphisms.

Higher Principle of the Indiscernibility

of Maximally Stretched Equals

The intrinsic properties of an object in a higher category

have to be invariant under

the maximally stretched notion of equality available.

The Category of Categories

• Since in category theory, any mathematical object is an object of a category, categories themselves have to be objects of a category.

• In order to define such a *category of categories Cat*, we need a notion of morphism between categories.

• Such a notion exists, namely *functors:* $F : \mathscr{C} \to \mathscr{D}$ given by

.maps on objects $X \mapsto F(X)$
.maps on morphisms

$$(f : X \to Y) \mapsto F(f) : F(X) \to F(Y) \text{ (covariant functor)}$$
$$(f : X \to Y) \mapsto F(f) : F(Y) \to F(X) \text{ (cotravariant functor)}$$

such that

.it respects identities: $F(id_X) = id_{F(X)}$
.it respects compositions $F(g \circ f) = F(g) \circ F(f)$

Isomorphism of Categories

- Two categories $\mathscr{C}$ and $\mathscr{D}$ are *isomorphic* if there are functors

$$F : \mathscr{C} \to \mathscr{D} \qquad \text{and} \qquad \mathscr{C} \to \mathscr{D} : F^{-1}$$

such that

$$F^{-1} \circ F = id_{\mathscr{C}} \qquad \text{and} \qquad F \circ F^{-1} = id_{\mathscr{D}}$$

where $id_{\mathscr{C}} : \mathscr{C} \to \mathscr{C}$ and $id_{\mathscr{D}} : \mathscr{D} \to \mathscr{D}$ are the corresponding identity functors.

- On objects, this definition takes the form:

$$(F^{-1} \circ F)(c) = c \qquad \text{for } c \text{ and object in } \mathscr{C}$$

$$(F \circ F^{-1})(d) = d \qquad \text{for } d \text{ and object in } \mathscr{D}$$

- This definition of isomorphism of categories uses a strict equality between functors.

Morphisms in *Cat*

- In general, the category of n-categories is a $(n + 1)$-category.

- In particular, the category *Cat* of 1-categories is a 2-category:

.Objects: 1-categories.

.1-morphisms: functors btw 1-categories.

.2-morphisms: ***natural equivalences*** btw functors.

Natural Equivalences

• Given two functors $F, G : \mathscr{C} \to \mathscr{D}$ a natural transformation is a 2-morphism in *Cat*

$$\eta : F \Rightarrow G$$

defined by a collection of maps

$$\eta_x : F(x) \to G(x) \qquad \text{(one for each } x \in \mathscr{C})$$

such that the following diagram commutes

$$\begin{array}{ccc} F(x) & \xrightarrow{\eta_x} & G(x) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(y) & \xrightarrow[\eta_y]{} & G(y) \end{array}$$

for every morphism $f : x \to y$ in $\mathscr{C}$

Natural Isomorphisms

• Since we have 2-morphisms, *isomorphisms of categories* is not the maximally stretched notion of equality in *Cat*.

• The maximally stretched notion of equality between categories is given by the so-called ***equivalence of categories***, where now

$F^{-1} \circ F = id_{\mathscr{C}}$ is stretched to a *natural isomorphisms* $F^{-1} \circ F \simeq id_{\mathscr{C}}$

• On objects, this stretching takes the form:

$(F^{-1} \circ F)(c) = c$ is stretched to an *isomorphism* $(F^{-1} \circ F)(c) \simeq c$

Categories are not Structured Sets

• Two categories with a different number of underlying objects might be equivalent.

• For instance,

.a category $\mathscr{C}$ with a unique object and trivial self-identity,

.a category $\mathscr{D}$ with two isomorphic objects

are equivalent, even if they have different underlying sets of objects.

• Hence, a category cannot be understood as a *structured set* since the underlying set of objects is not an intrinsic property, but just an artifact of the presentations.

Summary up to now

- **Set Theory**:

  .Mathematical objects:

  Sets + Structure + Laws

  .The unique available notion of equality is strict equality.

- **(Higher) Category Theory**:

  .The Stretching of Equality:

  $=$  $\simeq$ all the way up

  .Higher Principle of the Indiscernibility of Maximally Stretched Equals.

Main Thesis of Homotopy Type Theory

HoTT also postulates the existence of a fund. math. object, namely

∞-groupoids

which are a higher dimensional generalization of sets

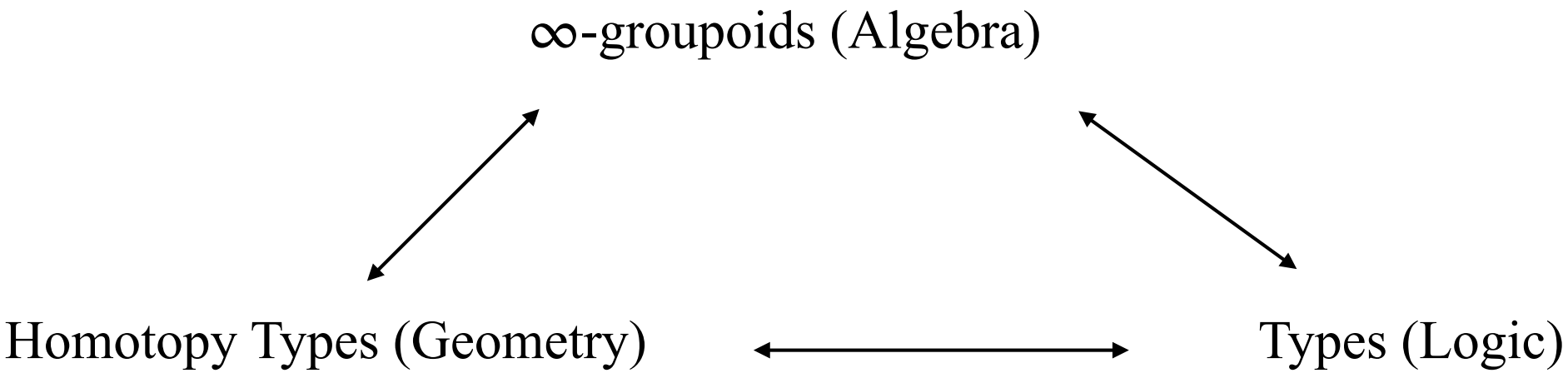
that incorporate the maximal stretching of strict equalities.

An ∞-groupoid is a higher category

in which every *n*-morphism

is invertible in the weakest sense of the term.

The HoTT Trinity



Mathematics is not built on top of a ***set-theoretic dust***,

but rather on a kind of ***topological foam***

made of ***abstract shapes***.

« [...] after Cantor and Bourbaki, no matter what we say, set theoretic mathematics resides in our brains. When I first start talking about something, I explain it in terms of Bourbaki-like structures: topological spaces, linear spaces, the field of real numbers, finite algebraic extensions, fundamental groups. I cannot do otherwise. If I'm thinking of something completely new, I say that it is a set with such-and-such a structure [...] That is, at first we start with the discrete sets of Cantor, upon which we impose something more in the style of Bourbaki. But fundamental psychological changes also occur. [...] Instead of sets, clouds of discrete elements, we envisage some sorts of vague spaces, which can be very severely deformed, mapped one to another, and all the while the specific space is not important, but only the space up to deformation. If we really want to return to discrete objects, we see continuous components, the pieces whose form or even dimension does not matter. Earlier, all these spaces were thought of as Cantor sets with topology, their maps were Cantor maps, some of them were homotopies that should have been factored out, and so on. I am pretty strongly convinced that there is an ongoing reversal in the collective consciousness of mathematicians: the right hemispherical and homotopical picture of the world becomes the basic intuition, and if you want to get a discrete set, then you pass to the set of connected components of a space defined only up to homotopy. That is, the Cantor points become continuous components […] »

Y. Manin in M. Gelfand, *We Do Not Choose Mathematics as Our Profession, It Chooses Us: Interview with Yuri Manin*, Notices of the AMS, 56(10).

« Badiou considère donc la théorie des ensembles comme un absolu. On peut se poser la question de savoir ce qu'apporte ce qualificatif d'« absolu » à la théorie des ensembles ? Ensuite cette théorie, qui a incontestablement dominé les mathématiques du 20e siècle, jouera-t-elle le même rôle au siècle suivant ? Cette théorie est basée sur la notion d'égalité ($a = b$ ou $a \neq b$) et sur la notion d'appartenance ($a \in A$, l'élément a appartient à l'ensemble A). Mais déjà la théorie des catégories (apparue dans les années 1950) a montré la nécessité d'affaiblir la notion d'égalité en celle d'« isomorphisme » et la théorie (en gestation) des ∞-catégories va encore plus loin. Dans cette nouvelle théorie, les égalités existent encore mais sont le niveau 0 d'une suite infinie de relations de plus en plus faibles, les isomorphismes occupant le niveau 1. Par exemple, sur une partie A du plan, au lieu de considérer la relation $a = b$ entre deux points a et b, on a une relation plus faible, à savoir qu'il existe un chemin continu dans $a = b$ qui va de a à b et étant donnés deux tels chemins, on peut se demander s'il existe une déformation continue de l'un dans l'autre, une deformation de ces déformations, etc. C'est la théorie de l'homotopie. Cette idée que des propriétés ne sont satisfaites qu'à homotopie près est en train de gagner la logique mathématique. Vladimir Voevodsky a le projet, pour le moins ambitieux, de remplacer la vieille théorie des ensembles ZFC par une nouvelle théorie dans laquelle, à la notion d'égalité, on substituerait celle d'égalité à homotopie près : c'est l'axiome d'univalence. Dans ce nouveau cadre, les ∞-catégories deviendraient des catégories usuelles, mais dans un contexte ensembliste totalement nouveau. C'est une révolution conceptuelle majeure qui est en préparation. »

P. Schapira, *L'Éloge des mathématiques d'Alain Badiou Ou le fantasme d'une théorie absolue.*

The Dialectics of Identity and Difference

« Arguably, nowhere else is the dialectic of identity and difference so rich and visible as in mathematics. […] Mathematicians keep affirming just that: that A equals B. No one else – not even poets – affirms such claims as often. The presence of the dialectic of identity and difference in mathematics is far from accidental: it is, quite simply, what mathematics is about. »

R. Netz, *The Aesthetics of Mathematics: A Study.*

Dialectics

(in the Hegelian sense of the term)

is

the rational process through which we can progressively sublate

the oppositions of the understanding.

(e.g. Identity and Difference, Finite and Infinite, Being and Nothingness, Good and Evil, Life and Death, Day and Night, Male and Female, Black and White, Love and Hate, Freedom and Necessity).

The notion of ∞-*groupoid* emerges as a result of

the dialectical sublation of the initial sharp opposition

Identity  Difference

encoded in *set theory*.

Set theory relies on a sharp opposition

Identity  *Difference*

« Two » sets A and B are either strictly equal (i.e. one and the same set)

if they have the same elements (***axiom of extensionality***)

$$A = B$$

(In this case, A and B are just different names of the same set)

or merely different

$$A \neq B$$

The set-theoretic conception of identity and difference

is defined by a sharp opposition between

A purely internal and trivial self-identity to the exclusion of difference

encoded in the ***Principle of Identity*** $A = A$:

Every thing is trivially identical to itself

and

A purely external difference

encoded in Leibniz's ***Principle of the Identity of Indiscernibles:***

Two numerically different things cannot be (qualitatively) equal.

Leibniz's Principle of the Identity of Indiscernibles

« There is no such thing as two individuals indiscernible from each other. »

G.W. Leibniz, *Letter to Clarke.*

« [...] in nature, there cannot be two individual things that differ in number alone. For it certainly must be possible to explain why they are different, and that explanation must derive from some difference they contain. [...] never do we find two eggs or two leaves or two blades of grass in a garden that are perfectly similar. »

G.W. Leibniz, *Primary Truths.*

No two things are perfectly alike, i.e. there are no perfect repetitions.

The *numerically distinct* is *qualitatively dissimilar*.

There are no differences « *solo numero* »

A world satisfying Leibniz's principle would exhibit a ***maximal qualitative diversity***.

Principle of Identity

There is no possible « equality »

between numerically different things

Each entity a is only equal to itself

$$a = a$$

and this exhausts

the meaning of the symbol equal $=$.

The Principle of Identity according to Hegel

« *The principle of identity accordingly reads: 'Everything is identical with itself; A = A'* […] This principle, instead of being a true law of thinking, is nothing but the law of the *abstract understanding*. […] If one maintains that this sentence cannot be proven but that each consciousness proceeds in accord with it and experientially concurs with it as soon as it hears it, then it is necessary to note, in opposition to this alleged experience of the school, the general experience that no consciousness thinks, has representations, and so forth, or speaks according to this law, that no concrete existence of any sort exists according to this law. Speaking according to this alleged law of truth ('a planet is – a planet', 'magnetism is – magnetism', 'the spirit is – a spirit') is considered, quite correctly, to be silly; this is presumably a universal experience. The school in which alone such laws are valid has, along with its logic which seriously propounds them, long since been discredited in the eyes of healthy common sense and in the eyes of reason. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline. Part I: Science of Logic*, § 115.

« […] to say of two things that they are identical is nonsense,

and to say of one thing that it is identical with itself is to say nothing at all. »

L. Wittgenstein, *Tractatus Logico-Philosophicus*, §5.5303

The classical theory of identity and difference according to Hegel

« Difference is [...] the diversity in which each of what is differentiated is for itself what it is and indifferent to its relation to the other which is thus a relation external to it. Because of the indifference of the diverse [things] to their difference, that difference falls outside them [...] Diversity has likewise been transformed into […] the principle that everything is diverse or that there are no two things that are completely like one another. Here 'everything' is provided with a predicate that is the opposite of the identity attributed to it in the […] principle [of identity] […] insofar as diversity pertains only to the external comparison, something is supposed to be only identical with itself for itself and thus this second principle [the principle of the identity of the indiscernibles] is supposed not to contradict the first [the principle of identity]. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline. Part I: Science of Logic*, § 117.

« [...] the identity that stays distant from difference, and the difference that stays distant from identity, are the products of external reflection and of an abstraction that arbitrarily clings to this point of indifferent difference. »

G.W.F. Hegel, *The Science of Logic*.

The Moment of the Understanding

or Vishnu, the Preserver, the Loving Kindness of God

- This is the moment in which thinking posits binary sharp oppositions

$$A \nleftrightarrow B$$

btw self-standing terms that entertain a relation of « indifferent difference » and remains in the « abstract *either-or* ».

- The opposite terms are isolated, one-sided, well-determined, and self-standing, i.e. they are not affected by the opposition.

- God kindly provides what each term requires in order to persist in its one-sided existence.

« Thinking as *understanding* does not budge beyond the firm determinateness [of what is entertained] and its distinctness over against others. […] The understanding is thus to be regarded as analogous to what one calls the *loving kindness* of God, insofar as we understand by this that finite things *are*, that they have a standing. Thus, for instance, in nature one recognizes the loving kindness of God in that the diverse classes and genera of both animals and plants have been endowed with everything they need in order to preserve themselves and flourish. […] To do philosophy, it is above all required that each thought be grasped in its full precision and that one is not content with vagueness and indeterminacy. It also, however, tends to be said that the understanding must not go too far. This is correct, insofar as the point of view of the understanding is not something ultimate [...] It is the way of youth to relish abstractions, whereas a person with the experience of life does not indulge in the abstract *either-or*, clinging instead to what is concrete. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline. Part I: Science of Logic*, §80.

The Dialectical Moment or Shiva, the Power of God

- The fixed and abstract determinations start to be alienated by the opposition.

- Each term self-sublates its own one-sidedness by passing into the opposite term and by introjecting the opposite term:

$$A_B \leftrightsquigarrow {}_A B$$

« The *dialectical* moment is the self-sublation of such finite determinations by themselves and their transition into their opposites. […] a process of going beyond the isolated determinacy […] [The principle of the dialectic] corresponds to the representation of God's *power*. We say that all things (i.e. everything finite as such) come to judgment, and with this we have a view of the dialectic as the universal, irresistible power which nothing, however secure and firm it may feel itself to be, can withstand. To be sure, the depth of the divine being, God's concept, is not yet exhausted by this determination. Still, it forms an essential moment in all religious consciousness. […] dialectic. It is the same principle that forms the basis of all other processes in nature and through which nature is at the same time driven beyond itself. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline. Part I: Science of Logic*, §81.

The Speculative Moment

or Brahma, the mystical *Coincidentia Oppositorum*

- The injecting of each term into its opposite is continued all the way down:

$$A_{B_{A_{B_{A_{B_{A\ldots}}}}}} \quad \leftrightsquigarrow \quad {}_{B}{}^{A}{}^{B}{}^{A}{}^{B}{}^{A}{}^{B}A$$

- The speculative moment encodes the ultimate instance that emerges when we continue the dialectical sublation of the initial sharp opposition all the way down.

- In this projective overarching instance, the initial opposite terms are inextricably entangled.

The Rational = The Mystical

« [T]he mystical is […] the concrete unity of those determinations that count as true for the understanding only in their separation and opposition. […] the abstract thinking of the understanding is so far from being something firm and ultimate that, to the contrary, it turns out to be constantly sublating itself and changing over into its opposite, whereas the rational as such consists precisely in containing the opposites as ideal moments within itself. Thus, everything rational is to be called at the same time 'mystical', by which, however, nothing more or less is said than that it goes beyond the understanding and in no way that it is to be regarded generally as inaccessible to thinking and as incomprehensible. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline. Part I: Science of Logic*, §82.

Equivalence Relations

Identity $\leftrightsquigarrow$ Identity Difference

This amounts to introduce

some form of « equality » (called *equivalence*)

between numerically different entities

that encode partial similarities.

The idea that any two things are just different

is too poor to ground rational thinking.

To think is among other to *abstract*,

that is to « forget » certain differences

that might be irrelevant in a given context.

By doing so, we can

classify,
recognize patterns, structures, regularities, forms,
extract invariants,
generalize,
engage in exchange process (linguistic, economic, etc.).

« Loss of memory as the transcendental condition of science.

All reification is forgetting »

M. Horkheimer and T.W. Adorno, *Dialectic of Enlightenment*, p. 191.

« Locke, in the seventeenth century, postulated (and rejected) an impossible idiom in which each individual object, each stone, each bird and branch had an individual name; Funes had once projected an analogous idiom, but he had renounced it as being too general, too ambiguous. In effect, Funes not only remembered every leaf on every tree of every wood, but even every one of the times he had perceived or imagined it. [...] He was, let us not forget, almost incapable of general, platonic ideas. It was not only difficult for him to understand that the generic term dog embraced so many unlike specimens of differing sizes and different forms; he was disturbed by the fact that a dog at three-fourteen (seen in profile) should have the same name as the dog at three- fifteen (seen from the front). [...] He was the solitary and lucid spectator of a multiform world which was instantaneously and almost intolerably exact. [...] I suspect, nevertheless, that he was not very capable of thought. To think is to forget a difference, to generalize, to abstract. In the overly replete world of Funes there were nothing but details, almost contiguous details. »

J.L. Borges, *Funes, The Memorious*.

Examples of similarities or equivalences

- Parallel lines in an Euclidean Plane.
- The arithmetical formulas $3 + 1$ and 2×2.
- Sets with the same number of elements.
- Frequencies 440Hz and 880Hz.
- Trains at 8h45am from Geneva to Paris (Saussure, *Course in General Linguistics*).
- *x* quarter corn = *y* quintal iron (Marx, *Capital*, I.1.1).
- A circle and a square (since both have a « circular » hole and the same number of points).
- A circle and a cylinder (since both have a « circular » hole).

In all these examples, we want to capture the similarities

by making abstraction of the corresponding differences.

From a mathematical standpoint, a process of abstraction

can be encoded in an ***equivalence relation*** $\sim$

on the initial domain of « concrete » individuals X.

The equivalence relation $\sim$

encodes the fact that certain individuals in X

will be considered equivalent in some respect,

which is denoted $a \sim b$.

Equivalence Relations

- **Relation** in a set X:

  .collection of oriented arrows $x \to y$ (possible including self-arrows $x \to x$) with at most one arrow between any two elements.

- **Equivalence relation** in a set X: relation (denoted $\sim$) such that:

  .$x \sim x$ (reflexivity),

  .$x \sim y \Rightarrow y \sim x$ (symmetry),

  .$(x \sim y)$ and $(y \sim z)$ implies $(x \sim z)$ (transitivity)

- The pair $(X, \sim)$ is called a *setoid*.

Quotient Set

- An equivalence relation $\sim$ induces a partition of X into subgroups called ***equivalence classes***, where x and y are in the same class if $x \sim y$.

- The ***set of equivalence classes*** is called ***quotient set*** and denoted

$$X/\sim$$

- The quotient map

$$q : X \to X/\sim$$

sends each $x \in X$ to its equivalence class $[x]$ such that

$$x \sim y \Rightarrow [x] = [y]$$

The fundamental importance

of equivalence relations in mathematics

relies on the fact that they encode

an essential operation of rational cognition,

namely *abstraction*.

Equivalence Relations Constitute *Abstracta*

- $3 + 1$ and 2×2 are *equi-valent* in the sense that they compute to the same *value*.

- In general, an equi-valence relation $a \sim b$ encodes the fact that a and b have the same generalized kind of « value » called the *abstract notion* defined by $\sim$.

- Each class $[x]$ defines a particular « value » of the corresponding *abstract notion*.

A « definition by abstraction »

based on the use of equivalence relations

amounts to transform

« a common feature into an ideal object »

(H. Weyl, *Philosophy of Mathematics and Natural Science*)

Fregean Abstraction Principle

Fregean abstraction transforms equivalent terms

in abstracta that are strictly identical:

$$a \sim b \Rightarrow [a] = [b],$$

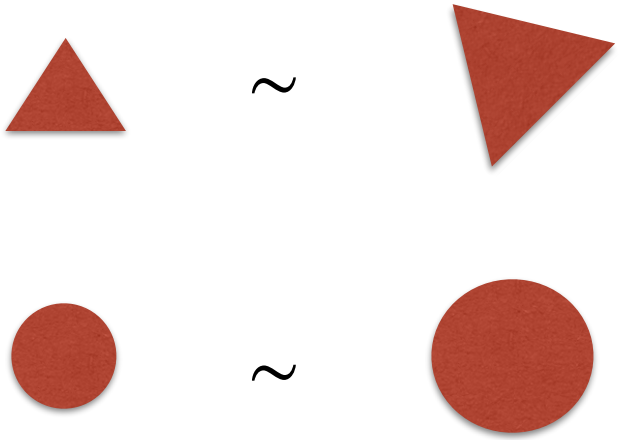
which means that $[a]$ and $[b]$ are just different presentations

of one and the same abstractum.

Equivalence relation given by « $a \sim b$ if they are similar »

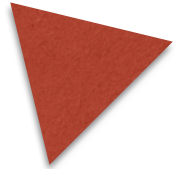
(Similarity: displacement + rescaling)

This equivalent relation makes abstraction from the position and the size.



$\Rightarrow$

Abstraction

Equilateral triangular shape $=$ {  ,  }

Round shape $=$ {  ,  }

$a \sim b \Rightarrow$ Shape (a) = Shape (b)

« If the relation is, e.g., that of geometrical similarity, then '*a* is similar to *b*' can be replaced by saying 'the shape of *a* is [strictly] the same as the shape of *b*'. »

G. Frege, *Letter to Russell, 28th July, 1902*.

Equivalence Relation	Abstractum
Computing to the same value	Value
Being parallel	Direction
Having the same number of elements	Cardinal number
Being *n* octaves apart	Chroma or pitch class
Being produced by the same amount of abstract labour	Exchange value
Being homeomorphic	Topological space
Being homotopically equivalent	Homotopy type (abstract shape)

Equivalence Relations as Groupoids

- An equivalence relation $\sim$ on a set X defines a category $\mathscr{C}_{\sim}$ such that

  .each $x \in X$ defines an object of $\mathscr{C}_{\sim}$,
  .there is a morphism btw x and y only if $x \sim y$.

  .Reflexivity of $\sim$: each x is equipped with a trivial self-identity id_x.
  .Transitivity of $\sim$: morphisms can be composed.

  .Symmetry of $\sim$: each $f : x \to y$ is reversible, i.e. an isomorphism.

- A category in which all morphisms are isomorphisms is called *groupoid*.

All in all, equivalence relation $\sim$ on a set X defines a groupoid $\mathscr{C}_{\sim}$ with many objects and one isomorphism between any two objects at most.

On Abstracta and Their Presentations

- In some cases the abstract process formally defines abstracta that might be beyond the realm of the available forms of intuition.

- For instance,

Homotopy type = an equivalence class of topological spaces.

- A homotopy type is a kind of ***abstract shape*** presented by different topological spaces, but it is not itself a space.

- In particular, a *punctured plane*, a *cylinder*, a *Möbius band*, and a *circle* are all presentations of the same homotopy type.

- Here we can

  .either develop forms of intuition adapted to the new abstracta and work directly with the latter,

  .or work with their presentations.

- In the last case, we have to distinguish

  .the presentation-dependent properties

  .the properties invariant under equivalences.

- This constraint is an instance of the *Principle of the Indiscernibility of Isomorphic Objects*.

« In scientific practice the introduction of a concept thus abstracted expresses the intention of exclusively considering *invariant* properties and relations among the originally given objects. »

H. Weyl, *Philosophy of Mathematics and Natural Science*

Examples

- A 2-dimensional cylinder and a 1-dimensional circle are presentations of the same homotopy type.

Hence, dimension does not define an intrinsic property of homotopy types.

- A « category » with a unique trivially self-identical object and a « category » with two isomorphic objects are presentations of the same category.

Hence, the number of objects does not define an intrinsic property of categories.

Groups

$$\text{Identity}_{\text{Difference}} \leftrightsquigarrow \text{Difference}$$

The identity principle

$$a = a$$

- is ***universally true***, i.e. it is valid for any a:

« No entity without identity »

(W.V.O. Quine, *Speaking of Objects*)

- it is ***evident***:

It is a truth that needs no proof.

- it is ***empty***, i.e. it conveys no information about a:

It is an irrelevant truth.

« In its positive formulation, $a = a$, [the principle of identity] is at first no more than the expression of empty tautology. It is rightly said, therefore, that this law of thought is without content and that it leads nowhere. »

G.W.F. Hegel, *The Science of Logic*.

Now, how can we inject

some form of difference within the identity

and go beyond this irrelevant « empty tautology »?

Hint to « detrivialize » the Identity Principle

The identity principle $a = a$ is a *proposition*.

But, *what is a proposition?*

Propositions according to « Platonists »

• For a « Platonist », mathematical objects and true propositions exist in themselves, i.e. independently of the means through which we can gain access to them.

• In this framework, *a proposition P is a statement that is either true or false…*

… and a proof of *P* is not itself a mathematical object…

… but rather an epistemic supplement thanks to which a subject can know that *P* is true.

Propositions according to Constructivists

- According to the constructivist tradition (Kant, Kronecker, Wittgenstein, Brouwer, Heyting, Kolmogorov):

> « *To exist means to be constructed* […] A mathematical assertion affirms the fact that a certain mathematical construction has been effected. »
>
> A. Heyting, *Intuitionism. An Introduction*, 1956.

- In particular, the assertion *The proposition A is true* means that a proof of *A* has been constructed.

- In order to give proofs the dignity of mathematical objects, ***a proposition is defined as the collection of its proofs***.

Brouwer's program […] consisted in the investigation of mental mathematical construction as such, without reference to questions regarding the nature of the constructed objects, such as whether these objects exist independently of our knowledge of them. […] If 'to exist' does not mean 'to be constructed', it must have some metaphysical meaning. It cannot be the task of mathematics to investigate this meaning or to decide whether it is tenable or not. We have no objection against a mathematician privately admitting any metaphysical theory he likes, but Brouwer's program entails that we study mathematics as something simpler, more immediate than metaphysics. In the study of mental mathematical constructions 'to exist' must be synonymous with t'o be constructed'. [...] A mathematical assertion affirms the fact that a certain mathematical construction has been effected.

A. Heyting, *Intuitionism. An Introduction*, 1956, pp. 1-3.

A false proposition is just an empty collection

but a true proposition might include different proofs.

Whereas truth values are binary (true or false),

proofs are multiple.

Different proofs of the same proposition provide different

senses, *modes of presentation*, or *witnesses* of its truth,

and no proof of a proposition might exhaust its whole meaning.

Wittgenstein on Proofs

« Proof, one might say, does not merely shew that it is like this, but: how it is like this. It shows how 13+14 yield 27.[…] It might be said: that every proof, even of a proposition which has already been proved, is a contribution to mathematics? But why is it a contribution if its only point was to prove the proposition? Well, one can say: the new proof makes a new connexion. »

L. Wittgenstein, *Remarks on the Foundations of Mathematics*.

« In mathematics there are not, first, propositions that have sense by themselves and, second, a method to determine the truth or falsity of propositions; there is only a method, and what is called a proposition is only an abbreviated name for the method. »

F. Waismann, *Wittgenstein and the Vienna Circle: Conversations*, p.33.

Curry-Howard correspondence between *proofs* and *programs*

- A proposition (e.g. $5 + 3 = 8$) is a *task* and the proofs are programs that *perform* the task (e.g. obtaining the number 8 by adding 5 and 3).

- Now, there might be different programs that perform the same task.

- These programs might differ in relevant ways: the memory they use, the time they need to perform the task, etc.

- For a computer scientist, it is not true that the only thing that matters is that the task is performed. *How* it is performed is an essential aspect of her work.

- From a computer science perspective, proof-relevance is essential.

How can we prove an equality $a = b$?

By constructing an *equalization*

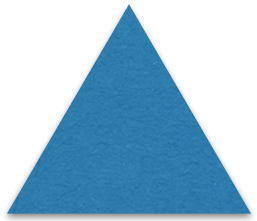
between a and b.

In particular, the principle of identity

$$a = a$$

can be proved by constructing a self-identification of a.

A triangle a can be « equalized with itself »



i.e. the proposition $a = a$ can be proved

- by not doing anything to it.

- by rotating it in 120° or 240°

- by reflecting it around the three principal axes.

These (composable and invertible) transformations defines the *group of symmetries* of the equilateral triangle D_3 (called *dihedral group*).

A *symmetry* is a transformation

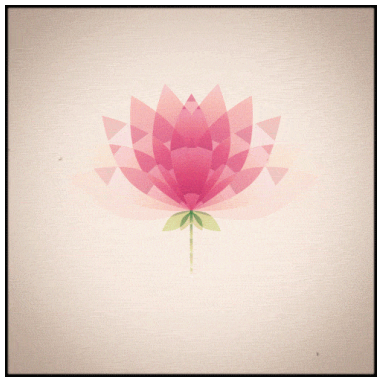
such that the final state cannot be distinguished from the initial one,

i.e. a non-trivial transformation that produces a trivial effect.

An object a with symmetries has a non-trivial identity,

in the sense that there are many proofs of $a = a$.

The symmetry group of an object defines its expanded identity.



« Sameness implies the relation of *with*, that is, a mediation, a connection, a synthesis: the unification into a unity. [...] But that unity is by no means the stale emptiness of that which, in itself without relation, persists in monotony. However, to get to the point where the relationship of the same with itself [...] emerges as this mediation [...] within identity, Western thought required more than two thousand years. For it is only the philosophy of speculative Idealism, prepared by Leibniz and Kant, that through Fichte, Schelling, and Hegel established an abode for the essence, in itself synthetic, of identity. [...] Just one thing we must keep in mind: since the era of speculative Idealism, it is no longer possible for thinking to represent the unity of identity as mere sameness, and to disregard the mediation that prevails in unity. Wherever this is done, identity is represented only in an abstract manner. »

M. Heidegger, *Identity and Difference*, p. 25.

On Objectivity and Automorphisms

The intrinsic (or objective) properties of an object still have to satisfy the

Principle of the Indiscernibility of

Isomorphic (in this case Automorphic) Objects.

« Objectivity means invariance with respect to the group of automorphisms. »

H. Weyl, *Symmetry*, 1952, p. 132.

The identity of an object imposes a constraint on its determinacy.

Example: Minkowski space-time

In Einstein's special theory of relativity,

(Minkowski) space-time is a highly symmetric object,

being its group of symmetries the so-called *Poincaré group*.

Hence, only the invariants under the action of the Poincaré group

define intrinsic properties of Minkowski space-time

(e.g. the temporal lapse between two events is not invariant).

On Symmetries, Resilience, and Inertia

Symmetric object

=

Object with a non-trivial identity

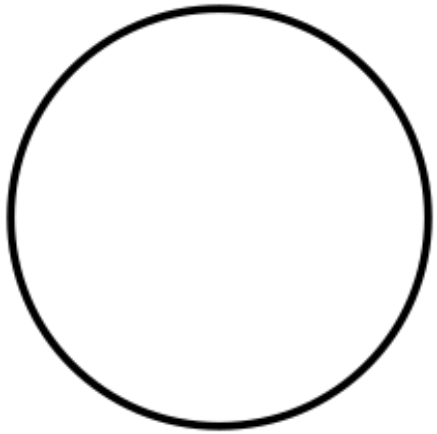
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***Plastic*, *resilient*, or *non-rigid* object**

=

« Heavy » object endowed with a of certain form of *inertia*.

For instance, a circle is « heavy »…



… since it has some « inertia » given by the fact

that certain transformations (rotations)

are automorphisms that do not change the circle as such.

An object with no symmetries has no « inertial resistance » to change

Groups as Groupoids

- A group $(G, \times, e)$ defines a category $\mathscr{C}_G$

  .with a unique element $\star$,

  .a morphism $\hat{g} : \star \to \star$ for each $g \in G$.

- The group operation guarantees that morphisms can be composed.

- The neutral element $e \in G$ defines the trivial identity $id_\star$.

- The existence of inverses in G guarantees that each $\hat{g} : \star \to \star$ is reversible, i.e. an isomorphism.

All in all, a group $(G, \times, e)$ defines a category $\mathscr{C}_G$

with a unique object $\star$ and many automorphisms $\hat{g}$ (one for each $g \in G$).

« Among the different ways of being identical to oneself, there is ‘inert’ identity, that is, the null action $id_{\star}$. What a group indicates is that this identity is but the degree zero of identity, its immobile figure and at rest. The other arrows of the group are dynamic identities. They are the active ways of being identical to oneself. »

A. Badiou, *Group, Category, Subject*.

Constructivist Principle of Identity $a = a$

- it is also universally true, i.e. it is valid for any a, since any a has at least the trivial group $G = \{e\}$ as group of symmetries:

No entity without identity (Quine)

- it might have non-trivial proofs given by the symmetries of a.

- it conveys information about a.

Plato on Degrees of Self-Similarity

« […] there is no shape more perfect

and none more similar to itself

[than the spherical shape] »

Plato, *Timaeus* (33b)

Current stage in the dialectization

of the sharp opposition between Identity and Difference:

$$\text{Identity}_{\text{Difference}} \leftrightsquigarrow \text{Identity}^{\text{Difference}}$$

Things are no longer merely different

since equivalence relations

inject a form of similarity into their difference.

Things are no longer trivially self-identical

since groups inject difference within their identity.

By just using 1-isomorphisms

we can still move one step further:

$$\textbf{Identity}_{\textbf{Difference}} \leftrightsquigarrow \textbf{Difference}^{\textbf{Identity}}_{\textbf{Difference}}$$

This yields ***groupoids,***

i.e. 1-categories in which every morphism is an isomorphism.

Groupoids

- A groupoid is a structure in which we can have both

  .many objects,

  .many isos between any two objects.

- Groupoids are the most general notion that we can construct with objects and 1-isomorphisms.

- **A groupoid is a notion that envelops**:

  .**sets** (many objects with trivial identities),

  .**groups** (a unique object with many isos),

  .**equivalence relations** (many objects with at most one iso).

Towards ∞-Groupoids

The next step in the dialectization

of the opposition between *Identity* and *Difference*

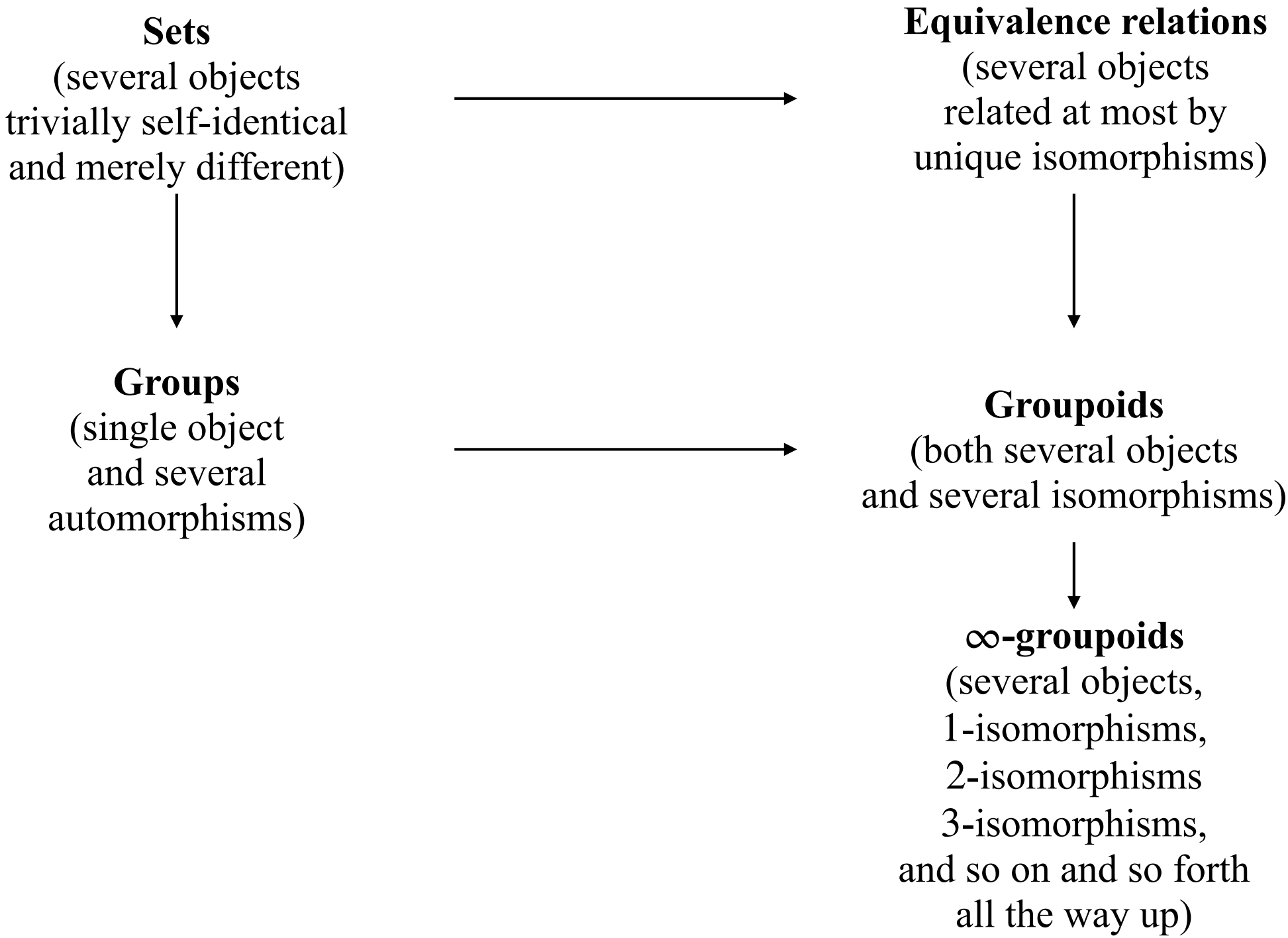
requires to introduce 2-isos between 1-morphisms.

• By pushing forward this process all the way down, we arrive to the notion of ∞ *-groupoid*.

• An ∞-groupoid is a structure in which we have:

.objects

.n-morphisms that are invertible in the weakest possible sense for every $n = 1,2,...,\infty$.


Sets
(several objects
trivially self-identical
and merely different)
Equivalence relations
(several objects
related at most by
unique isomorphisms)
Groups
(single object
and several
automorphisms)
Groupoids
(both several objects
and several isomorphisms)
∞-groupoids
(several objects,
1-isomorphisms,
2-isomorphisms
3-isomorphisms,
and so on and so forth
all the way up)

Type Theory

Homotopy Type Theory

- **Pre-history** (1900s):

  .Russell's theory of types,
  .Church's λ-calculus,
  .mathematical constructivism/intuitionism (Brouwer, Heyting, etc.).

- **Intuitionistic type theory**: P. Martin-Löf (1970s)

  Mathematical objects are *terms* of a *type* $a : X$
  endowed with a type-dependent notion of equality $=_X$

- **Homotopic interpretation**: Awodey-Warren (2005), Voevodsky (2006):

  Types are spaces and terms are points.

- **Univalence Axiom**: Voevodsky (2009):

  Types = Abstract Shapes (Homotopy Types).

- **Higher Inductive Types**: Shulman (2011).

Type theory is a fundamental logical system in which

the maximally stretched notion of equality provided by higher category theory

and the principle that « equal » objects have the same properties

are built into the system from the very beginning.

« Being is said in many ways »

(Aristotle, *Metaphysica*, 1003a33)

« No entity without type »

(P. Martin-Löf)

Each entity a is a case (a *term*)

of a particular kind or type of entities X,

denoted $a : X$.

Data Types in Computer Science

Data Type = specification of a type of data:

.of its range of possible values
.the operations one can perform on them.

Examples:

- String (plain text): Hello, 222
- Integer: 3, 99
- Floating-point (numbers with a decimal component): 3.14
- Boolean: 0,1

Kinds of Statements in Type Theory

- **Propositions**: statements with a truth value.

- **Judgements**: *speech* or *performatic acts* (like promises, nominations, marriages, etc.) without a truth value, like:

  $.a : X$ (declaration that the term a is of type X; no dual citizenship).

  $.A \equiv B$ (declaration that the expressions A and B are synonyms).

Type Constructors

Each type X is equipped with constructors that define its behavior:

- **Formation** of the corresponding type X, specifying under which conditions a type can be formed.

- **Introduction** of canonical terms $a : X$ of the type.

- **Elimination:** induction process based on the canonical terms.

The Empty Type $\mathbf{0}$

- **Formation:** there is a type $\mathbf{0}$.

- **Introduction:** no canonical terms.

- **Induction** (*ex falso quodlibet* or *principle of explosion*): if $\mathbf{0}$ had a term x, then any proposition P would be provable.

If there were something in nothingness,

then everything would be true.

- **Logical Meaning**: $\mathbf{0}$ is the *universal falsehood.* A proposition A is false if there is a function

$$A \to \mathbf{0}$$

(such a function exists only when A is empty, i.e. false)

- **Geometric Meaning**: $\mathbf{0}$ is the (-1)-sphere S^{-1}.

The Unity Type 1

• **Formation**: there is a type $\mathbf{1}$.

• **Introduction**: the type $\mathbf{1}$ has a unique canonical term

$$\star : \mathbf{1}$$

• **Elimination-Induction**: if $P(\star)$ holds and $x : \mathbf{1}$, then $P(x)$ also holds, i.e. all the terms have the same properties.

In particular, $\star = \star$ implies $\star = x$ for every $x : \mathbf{1}$, i.e. there is only one term *up to equality*.

• **Logical meaning**: $\mathbf{1}$ is the *universal truth*. A proposition A is true if there is a *truth incarnation* map « pointing » to a proof of A:

$$\mathbf{1} \to A$$

(if A is false, there are no terms to which we would send the terms in $\mathbf{1}$)

The Dyad Type 2

- **Formation:** there is a type **2**.
- **Introduction:**

$$\mathit{on} : \mathbf{2} \textbf{ and } \mathit{off} : \mathbf{2}$$

- **Induction**: …
  - **Geometric Meaning**: **2** can be understood as a 0-sphere S^0

    (the border of a 1-disk or interval).

(Note: this type is also called the *Booleans*)

The Type of Natural Numbers $\mathbb{N}$

- **Formation:** there is a type $\mathbb{N}$.

- **Introduction:**

  . $0_{\mathbb{N}} : \mathbb{N}$
  . $suc : \mathbb{N} \to \mathbb{N}$

This generates all the natural numbers:

. $1 \equiv suc(0_{\mathbb{N}})$
. $2 \equiv suc(suc(0_{\mathbb{N}}))$
. $3 \equiv suc(suc(suc(0_{\mathbb{N}})))$

- **Induction**: If $P(0)$ and $P(n) \to P(suc(n))$ hold, then $P(n)$ holds for every n.

Function Types

• **Formation:** given two types A and B we can form the function type

$$A \to B$$

• **Introduction:** every « functional form » $f(x)$ that yields a term $f(a) : B$ for each $a : A$ define a canonical term

$$f : A \to B.$$

(which does not mean that every term of $A \to B$ necessarily comes from a functional form).

• **Elimination (application):** given $f : A \to B$ and $a : A$, we can apply f to a, thereby obtaining a term $f(a) : B$.

Constructivist *Propositions-as-Types* Paradigm

In type theory, a proposition is not just

a statement that is either true or false (classical definition)

but rather a type P

whose terms

$$p : P$$

are its proofs.

Performatic act of proving $P \rightsquigarrow$ Construction of a term $p : P$

In type theory,

there is no truth without proof

Proofs are mathematical objects

(i.e. terms of a type)

that encode different

senses,

modes of presentations,

or *witnesses*

of the truth of the proposition.

Propositional Logic - Type Theory Dictionary

Propositions $\leftrightarrow$ Types

False $\leftrightarrow$ $\mathbf{0}$
(A is false if $A \to \mathbf{0}$ is inhabited)

True $\leftrightarrow$ $\mathbf{1}$
(A is true if $\mathbf{1} \to A$ is inhabited)

A and B $\leftrightarrow$ $A \times B$
(A and B is true there is a proof $a : A$ of A and a proof $b : B$ of B)

A or B $\leftrightarrow$ $A + B$
(A or B is true there is a proof $a : A$ of A or a proof $b : B$ of B)

A implies B $\leftrightarrow A \to B$
(A implies B if there exists a function $f : A \to B$
that sends proofs $a : A$ of A to proofs $f(a) : B$ of B)

Predicate Logic

• In predicate logic we do not only have propositions like

2 *is an even number* (endowed with a truth value),

but also predicates like

x is an even number (without a truth value).

• A predicate is a family of propositions, one for each possible value of the *free variable x.*

• A predicate acquires a truth value only when x is substituted with a constant :

Predicate $P(x)$ $\xrightarrow[x \mapsto a]{}$ Proposition $P(a)$

• In type theory, a predicate $P(x)$ is called a *family of types* or a *dependent type*, which means that for each $a : A$ we have a proposition $P(a)$.

Propositions $\longrightarrow$ Types
Predicates $\longrightarrow$ Families of types

Quantifiers

• Besides substitution, a predicate $P(x)$ can be converted into a proposition by « bounding » the variable x.

• Examples of bounding operators:

.**Existential quantifier**:

$\exists x$ such that P(x) is true

(e.g. *There exists x such that x is mortal*)

.**Universal quantifier**:

$\forall x$, $P(x)$ is true

(e.g. *For all x, x is mortal*)

Geometric Interpretation of Type Theory

- Type A ⇜⇝ Space.
- Term $a : A$ ⇜⇝ Point in the space.

In particular, a proposition A is the « space of reasons » of its truth,

i.e. a space such that each point $a : A$ is a proof of A.

- Predicates $P(x)$ for $x : A$ ⇜⇝ Fibrations over A.
- Existential quantifier $\exists x, P(x)$ ⇜⇝ Total space of the fibration, denoted $\Sigma_{(x:A)}P(x)$.

A term of this type is a pair $(a : A, p : P(a))$.

- Universal quantifier $\forall x : A, P(x)$ ⇜⇝ Space of *sections* of the fibration, denoted $\Pi_{(x:A)}P(x)$.

A term of this type is a *section* that sends each $a : A$ to a proof $p : P(a)$.

Type Theory	Homotopy Theory
Type	Space
Term	Point
Predicates	Fibrations
Existential quantifier	Total space of the fibration
Universal quantifier	Space of sections of the fibration

Type-Theoretic Equality

The notion of equality is

type-dependent,

i.e. *relative* to the type of the corresponding terms.

« But since things are said to be […] in many ways, the others that are said to be in accord with these must also follow along with them, so that *same*, *other*, and *contrary* must be distinct in each category. »

Aristotle, *Metaphysica*, 1018a35.

We shall now enrich the slogan

No entity without type (Martin-Löf)

with the slogan

No type without equality (Martin-Löf)

which will imply the slogan

No entity without identity (Quine)

Definition of Equality in Predicate Logic

- **Reflexivity** $a = a$

- **Principle of substitution *salva veritate***:

  Given a predicate in two variables $P(a, b)$, let's suppose that $a = b$. Then,

$$P(a, a) \Rightarrow P(a, b)$$

From this we can deduce:

- **Symmetry** : $(x = y) \Rightarrow (y = x)$

- **Transitivity**: $(x = y \text{ and } y = z) \Rightarrow x = z$

- **Principle of Indiscernibility of Equals**: equal things have the same properties (we take $P(x, y) \equiv Q(x) \Rightarrow Q(y)$, where $P(x, x)$ always holds. Hence, $x = y$ implies that $Q(x) \Rightarrow Q(y)$).

Formation Rule for the Equality Type

Given $a, b : X$, we can form the type

$$a =_X b.$$

Equality is a relative or type-dependent notion

A proposition stating that two terms of different types are equal

— like saying that a hippopotamus and the sinus function are equal —

is not even false, it is just ill-formed, i.e., meaningless.

Whereas a definitional equality $a \equiv b$

is a *judgement*, a *definition*, or a *stipulation* encoding *synonymy*

and with no true value,

a propositional equality $a =_X b$ is a *proposition*

that can be either true or false.

According to the propositions-as-types paradigm, a propositional equality

$$a =_X b$$

is the type of proofs that a and b are equal.

Such a proof $p : a =_X b$ will be called an *equalization* of a and b.

Equalizations = geometric *congruences*

« Things which coincide with one another [when superposed] are equal to one another »

Euclid's *Elements*, Notion 4.

« Two figures are congruent if one can be placed on the other without changing anything in the two figures except their place. »

G.W. Leibniz, *Letter to Christian Huygens* (1679).

« If indeed one tries to clarify the notion of *equality*, which is introduced right at the beginning of Geometry, one is led to say that two figures are equal when one can go from one to the other by a specific geometric operation, called a motion »

E. Cartan, *Notice sur les travaux scientifiques* (1974).

Two geometric figures a and b

in a space X are equal if there is a…



… that allows us to superpose a with b

(if we cannot do that, $a = b$ would be false).

***Equalizations* between terms/points are *paths* in the type/space**

If the terms $a, b : X$ can be interpreted as *points* in a *space*,

an equalization

$$\gamma : a =_X b$$

is a *path* between a and b:

$\gamma : [0,1] \to X$ such that $\gamma(0) = a$ and $\gamma(1) = b$

Equality ≡ Path-connectedness

Introduction Rule for the Equality Type

The self-identity type

$$a =_X a$$

is always inhabited by a canonical term

$$refl_a : a =_X a$$

encoding the trivial self-identification of a

In the homotopic semantics, $refl_a$ becomes the ***constant path*** based at a:

$$\gamma_a : [0,1] \to X \text{ such that } \gamma_a(t) = a, \forall t.$$

Self-identity might be non-trivial

The fact that there is always a canonical term $refl_a$ of $a =_X a$

does not mean that there are no other terms.

In HoTT, there might be non-trivial proofs

of the self-identity type

$$a =_X a$$

i.e proofs $\gamma : a =_X a$ different from $refl_a$.

Geometrically, such a proof $\gamma : a =_X a$

would be a non-constant loop based at a.

Now, the fact that there are non-trivial loops

$$\gamma : a =_X a$$

does not necessarily mean that γ

is propositionally different from $refl_a$.

Since $a =_X a$ is itself a type,

we can form the higher identity types

$\gamma =_{(a=_A a)} \eta$ where $\gamma, \eta : a =_X a.$

When two paths γ and η with the same endpoints in X are equal?

We can here apply the same Euclidean criterion of geometric congruence:

$\gamma =_{(a=_A a)} \eta$ is true

if we can go from γ to η by means of a continuous motion.

Such a path btw paths

with the same source and target

are called ***homotopies***.

Two paths γ and η that can be continuously deformed into one another

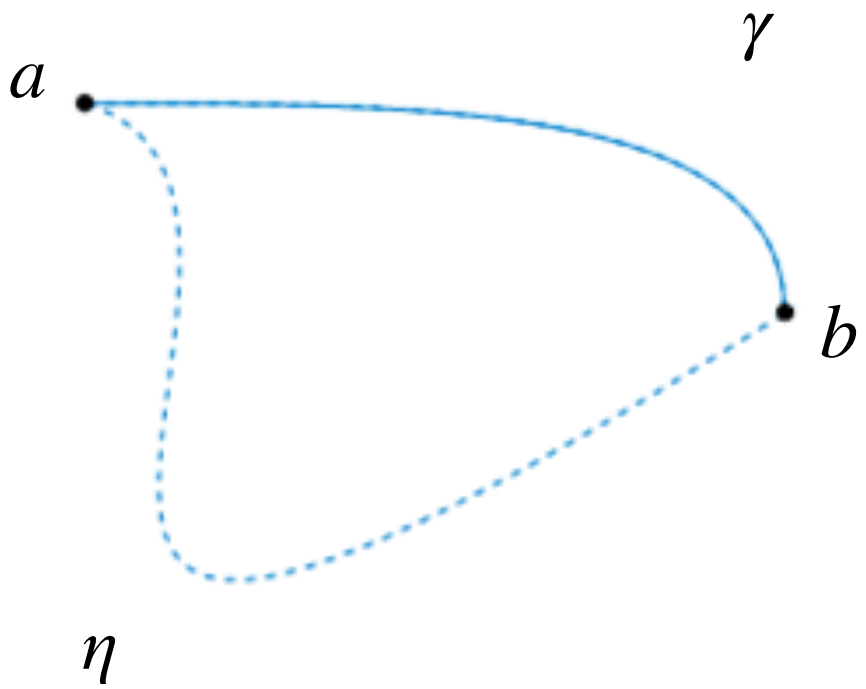
are called *homotopic*.

In general, given

$$\gamma, \eta : a =_X b,$$

a homotopy is a term

$$H : \gamma =_{(a =_X b)} \eta$$



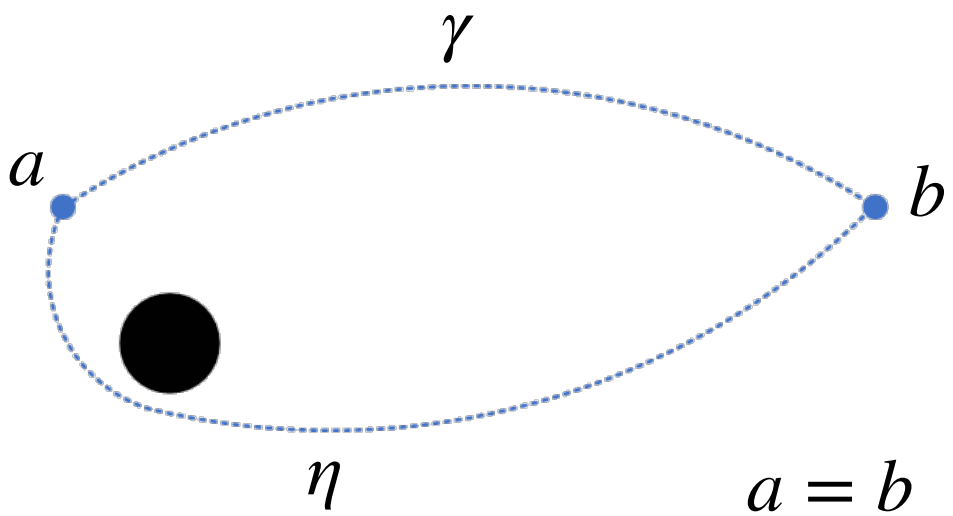
Now, if the space has holes,

there are paths that cannot

be mutually equalized,

i.e., paths that are not

homotopic.



$a = b$

$\gamma \neq \eta$

Holes are obstructions

to the equalizations between paths.

Is $\gamma =_{(a=_A a)} refl_a$ true or not?

In the homotopic interpretation,

$a : X$ is equipped with non-trivial self-identifications

(loops γ different from $refl_a$)

when the space X has ***holes***.

Indeed, a loop $\gamma : [0,1] \to X$ based at a

that surrounds a hole is not homotopic

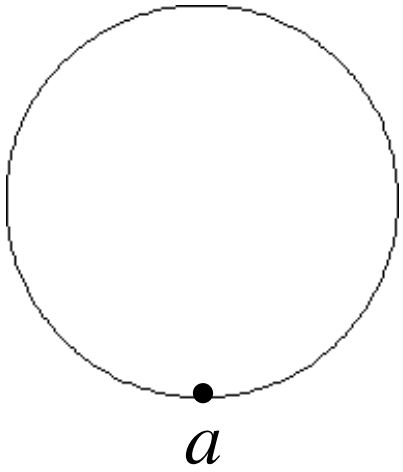
to the constant path $refl_a$.

« There are many ways of getting home; and one of them is to stay there. The others are to walk round the holes of the world till we come back to the same place. »

Pseudo-Chesterton , *The Everlasting Human*.

If the space has no holes,

every loop can be contracted to the trivial loop.



Without holes, identity is trivial.

Si l'*A* est tant *A* que ça, qu'il y reste !

Pourquoi le séparer de lui-même

pour si vite le rassembler?

J. Lacan, *Séminaire IX, L'identification*.

Homotopy Theory

Homotopy Theory

• ***Homotopy Theory*** is a branch of ***Algebraic Topology***.

• Algebraic Topology aims to establish *functorial bridges* between

Topological spaces ↔ Algebraic objects

by means of which we can

.use algebraic methods to study geometric objects

.« schematize » algebraic entities as spaces.

On *Form* and *Formalism* or the *Geometry-Algebra Duality*

Whereas geometric ***forms*** engage

our most intuitive, gestaltic, holistic, and higher-dimensional

rational faculties,

the consideration of ***formalism*** requires

full attention to the details

exhibited by the linear and discrete scriptural regime of signs.

« Algebra is just written geometry, geometry is just figurative algebra »

S. Germain, *Œuvres philosophiques*.

Conceptual History of Homotopy Theory

• First, Poincaré (1895) established a correspondence

Topological spaces $\rightarrow$ **fundamental groups** $\pi_1(X, a)$

Limitations:

.$\pi_1(X, a)$ only captures the 1-holes of X.
.$\pi_1(X, a)$ depends on a.

• The first limitation can be bypassed by using

Higher homotopy groups $\pi_n(X, a)$

• The second limitation can be bypassed by using:

Fundamental groupoids $\Pi_1(X)$

• All this can be summarized by using:

Fundamental ∞-groupoids $\Pi_\infty(X)$

Homotopy Theory as a Spaghetti-Western

- In order to catch holes in a space X we can throw a « lasso » to X and try to contract it.

- If we cannot contract it, it means that we have caught a hole in X.

- There might be holes that cannot be caught by throwing « lassos » of dimension 1.

- For instance, the 2-sphere S^2 has no *circular* holes.

- However, S^2 does have a hole, the hole within the sphere !!!

- Hence, higher dimensional holes can be caught by throwing generalized « lassos » given by spheres S^n of any dimension $n = 0,1,2,...$

n-spheres S^n

n-spheres S^n = n-dimensional boundaries of $(n+1)$-balls D^{n+1}

- Sequence of n-spheres S^n:

  .S^0: 0-dimensional border of a 1-dim. interval (set of two points).

  .S^1: 1-dimensional border of a 2-dim. ball (ordinary circle).

  .S^2: 2-dimensional border of a 3-dim. ball (ordinary sphere).

  .S^3: 3-dimensional border of a 4-dim. ball (hyper-sphere).

  .S^{-1} (backwards generalization): empty set $\varnothing$ (border of the 0-dim. ball, which is just a point).

Fundamental Group of a Space (I)

Given a space X and a *basepoint* a in X,

the fundamental group of X at a

$$\pi_1(X, a)$$

is given by the set of homotopy classes of loops in X based at a:

$$\gamma : [0,1] \to X \text{ such that } \gamma(0) \equiv a \text{ and } \gamma(1) \equiv a$$

- where the group operation is given by composition of paths,
- the neutral element is given by the constant path $\gamma_a(t) \equiv a$ for all $t \in [0,1]$.
- and the inverse of a path $\gamma(t)$ is given by the same graph of $\gamma(t)$ traversed in the opposite direction

$$\gamma(t)^{-1} \equiv \gamma(1-t)$$

The Fundamental Group of a Space (II)

- Let's consider the loops in a space X based at a. Since

  .loops can be concatenated $\gamma_1 \circ \gamma_2$,

  .there is a constant or trivial loop γ_a,

  .for each loop $\gamma : [0; 1] \to X$ there is a reverse loop

$$\gamma^{-1} = \gamma(1 - t)$$

  that traverses the path γ in the reversed sense,

it seems that we could define a group-like structure out of the loops based at a

where the concatenation would play the role of the group operation.

The Fundamental Group of a Space (III)

- However, this does not work since

  .concatenation is not associative:

$$(\gamma_0 \circ \gamma_1) \circ \gamma_2 \neq \gamma_0 \circ (\gamma_1 \circ \gamma_2)$$

  .the constant path does not act as an identity:

$$\gamma \circ \gamma_a \neq \gamma$$

  .the reverse path γ^{-1} is not an inverse:

$$\gamma \circ \gamma^{-1} \neq \gamma_a$$

- Now, in each of these expressions each side defines the same graph in *X,* but traversed at different speeds.

- Since there can be no hole obstructing their continuous deformations into one another, both sides in each expression are homotopic.

The Fundamental Group of a Space (IV)

• All these expressions that we need on order to define a group of loops hold if we make abstraction from the differences between homotopic loops.

• According to Fregean abstraction, we can forget these differences by considering equivalence classes $[\gamma]$ where $\gamma \sim \gamma'$ if they are homotopic.

• The set of homotopy classes of loops forms a group denoted

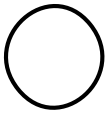
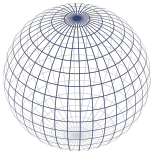
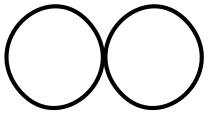
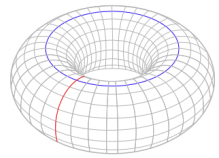
$$\pi_1(X, a)$$

and called ***fundamental group of*** X ***based at*** a (introduced by H. Poincaré around 1895).

• The fundamental group encodes the obstructions to the contractions of loops based at a, i.e. the 1-holes of the component of X that contains a.

• If the space has no 1-holes, then $\pi_1(X, a)$ is trivial, i.e. its unique element is $[\gamma_a]$.

What is the fundamental group of

- the Euclidean plane,
- a circle 
- a sphere 
- an eight (wedge product of two circles) 
- a torus (cartesian product of two circles) 

?

Fundamental Group of the Circle S^1

All the loops based at a in a circle S^1

are characterized by a ***winding number*** $n \in \mathbb{Z}$

that specifies how many times the loop encircles the hole.

- $n > 0$ means that the loop encircles the hole clockwise n times.
- $n = 0$ means that the loop does not encircle the hole.
- $n < 0$ means that the loop encircles the hole counter-clockwise n times.

$$\pi_1(S^1, a) \simeq \mathbb{Z}$$

Fundamental Group of the Eight $S^1 \vee S^1$

• Let's denote a the loops that winds clockwise around the right half of $S^1 \vee S^1$…

… and b the loops that winds counter-clockwise around the left half of $S^1 \vee S^1$.

• A loop at a basepoint b is defined by a sequence of turns around the right circle and around the left circle.

• An element in

$$\pi_1(S^1 \vee S^1, x_0)$$

is a « word » made of a sequence of letters a and b like for instance

$$aaa^{-1}bb^{-1}babb \qquad \text{which can be reduced to} \qquad abab^2$$

by cancelling segments of the form aa^{-1}, $a^{-1}a$, bb^{-1} and $b^{-1}b$.

• This group (the « free group in two letters ») is not commutative:

$$ab \neq ba$$

Fundamental Group of the Torus (I)

In particular, there are two classes of loops

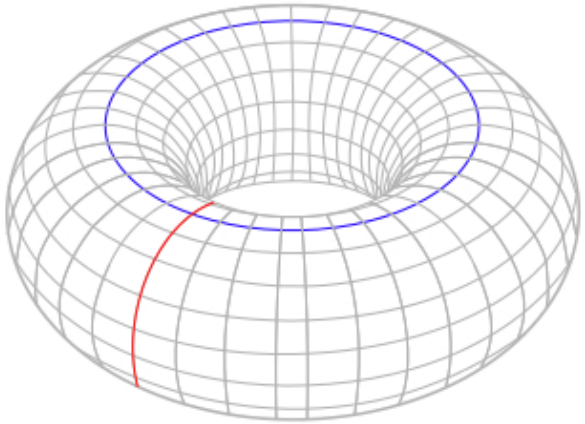
that cannot be deformed into the trivial loop,

the class of loops *a* that encircle the central hole of the torus

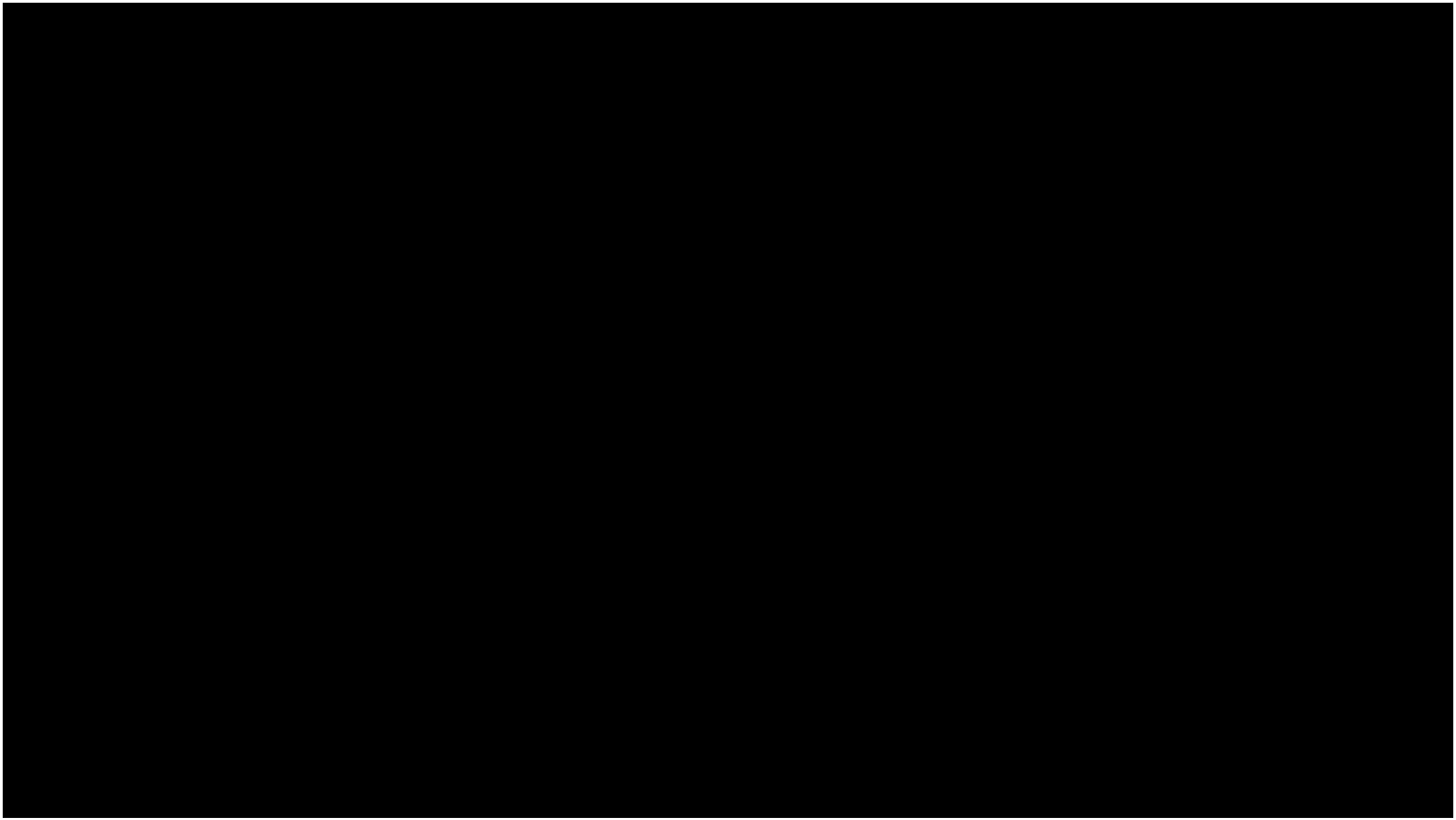
and the class of loops *b* that encircle the hole within the curved cylinder.

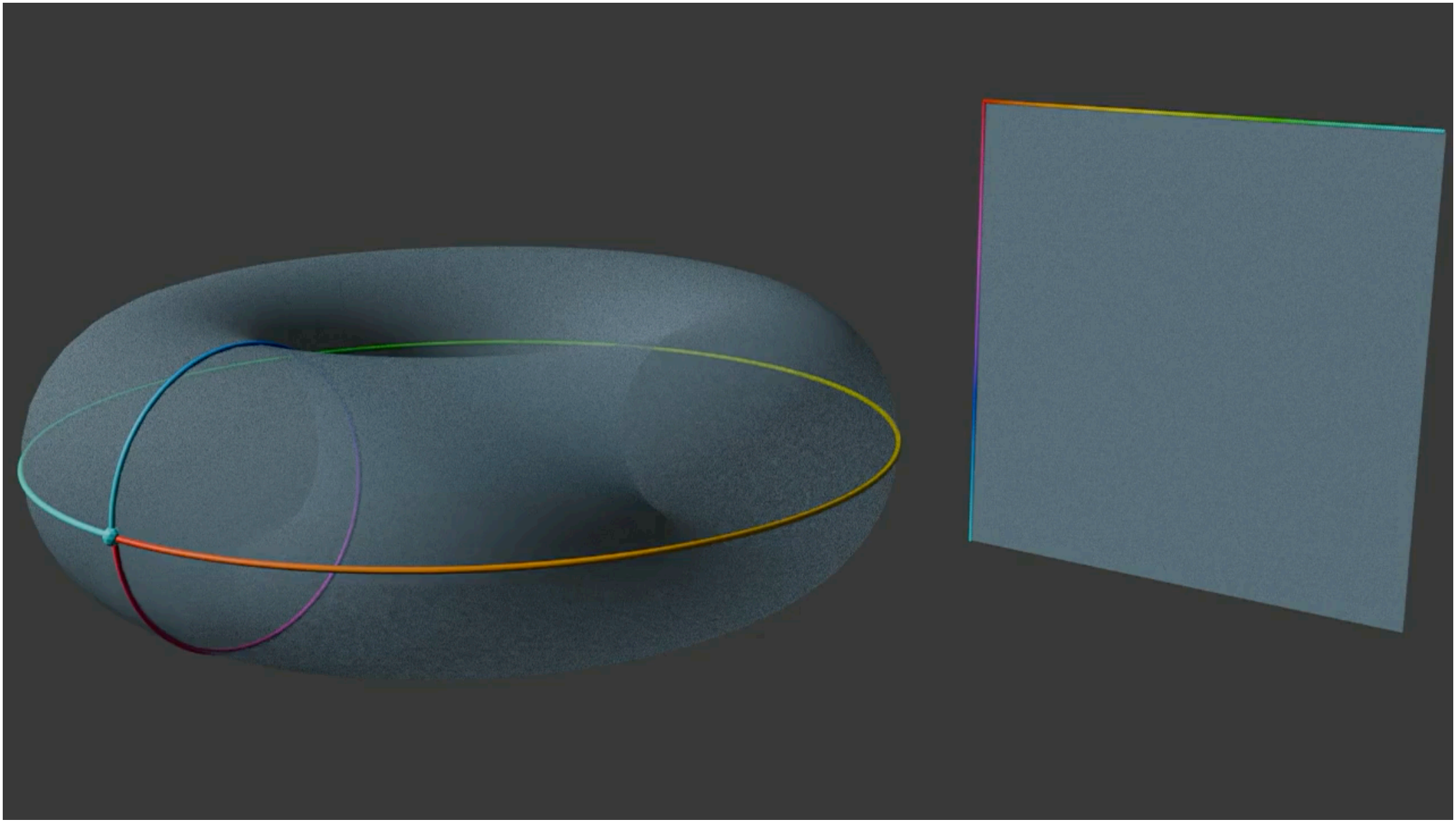
«… ces cercles qui font
le tour du trou du tore »

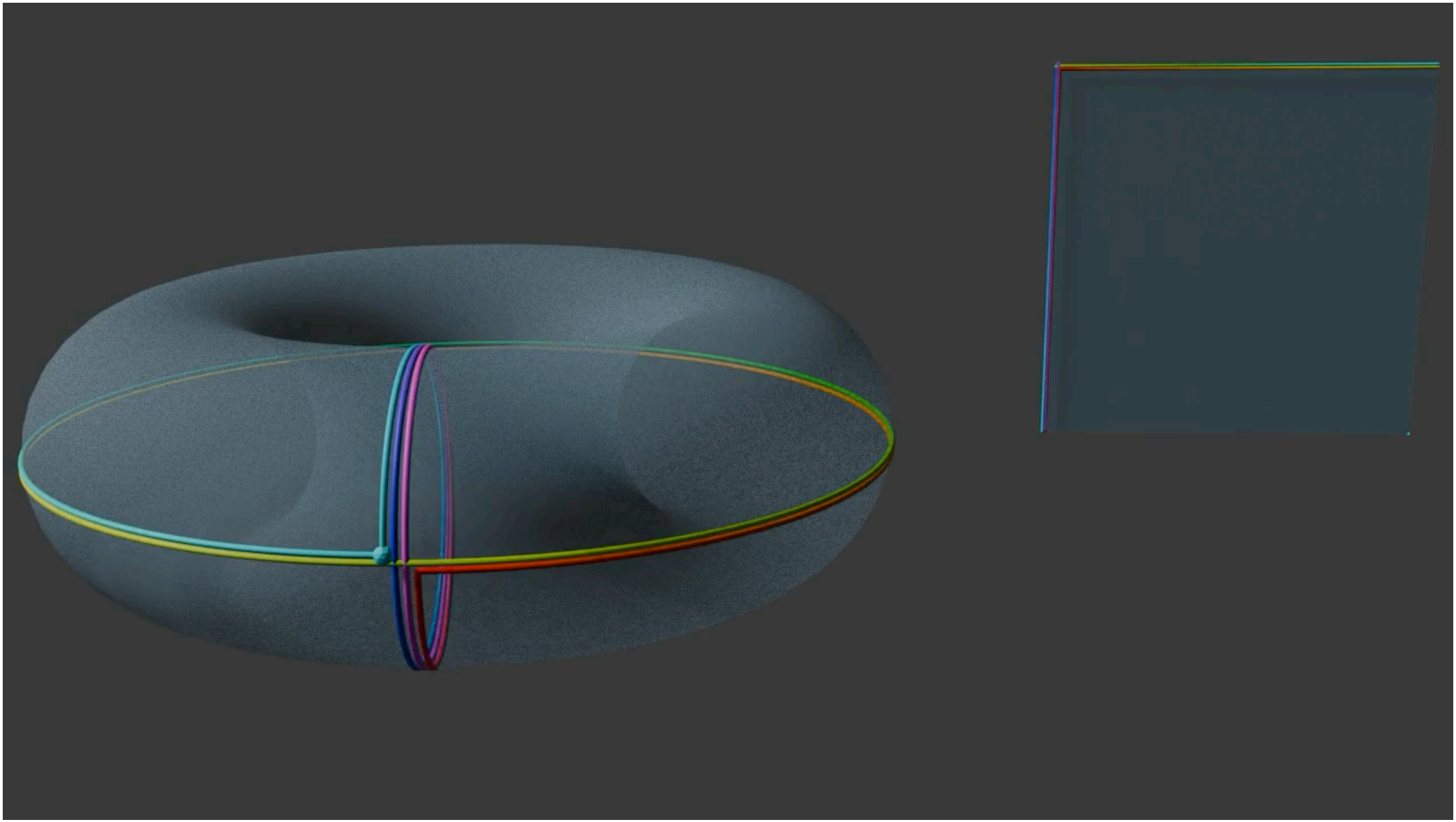
J. Lacan.



Torus

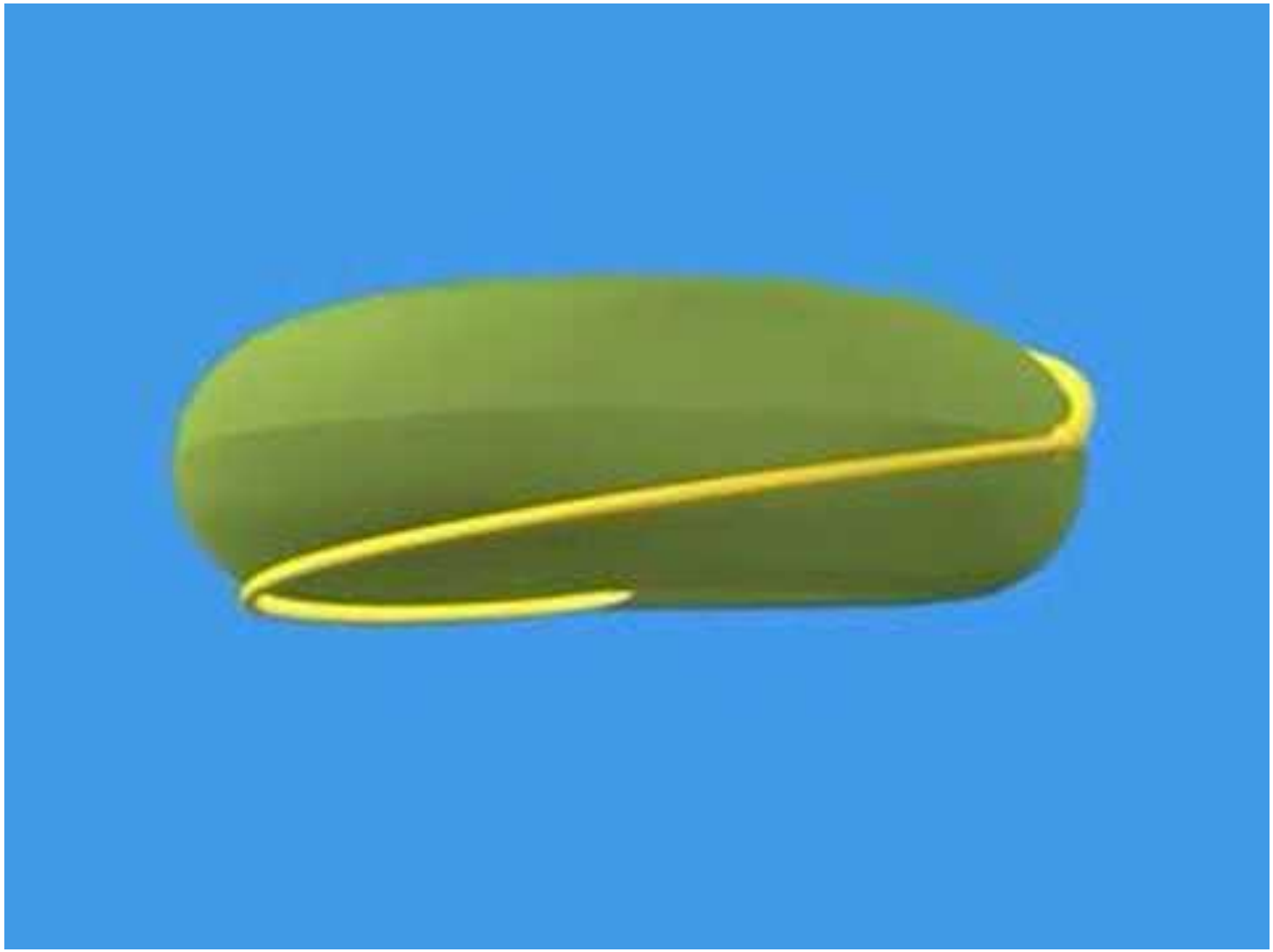






The Fundamental Group of the Torus is Commutative

$$ab = ba$$



The fundamental group of the torus T^2 is

$$\pi_1(T^2, b) = \mathbb{Z} \times \mathbb{Z}$$

An element in this group is a couple of natural numbers (n_a, n_b)

specifying the number n_a of turns

a that encircle the central hole of the torus

and the number n_b of turns b

that encircle the hole within the curved cylinder.

Towards Higher Homotopy Groups

• We have defined the fundamental group $\pi_1(X, a)$ by making abstraction of the differences between homotopic loops.

• So, this (Fregean) abstraction process relies on an operation of *straightening*, in the sense that we have replaced equivalent/ homotopic loops

$$l \sim l'$$

related by a stretched equality (the homotopies) with a strict equality

$$[l] = [l']$$

between the equivalence classes.

• By doing so, we have « truncated » the higher structure of homotopies, 2-homotopies between homotopies, etc.

Le nénuphar Blanc - Stéphane Mallarmé

« Résumer d'un regard la vierge absence éparse en cette solitude et, comme on cueille, en mémoire d'un site, l'un des ces magiques nénuphars clos qui y surgissent tout à coup, enveloppant de leur creuse blancheur un rien, fait de songes intacts, du bonheur qui n'aura pas lieu et de mon souffle ici retenu dans la peur d'une apparition, partir avec: tacitement, en déramant peu à peu sans du heurt briser l'illusion ni que le clapotis de la bulle visible d'écume enroulée à ma fuite ne jette aux pieds survenus de personne la ressemblance transparente du rapt de mon idéale fleur. »

The White Water Lily - Stéphane Mallarmé

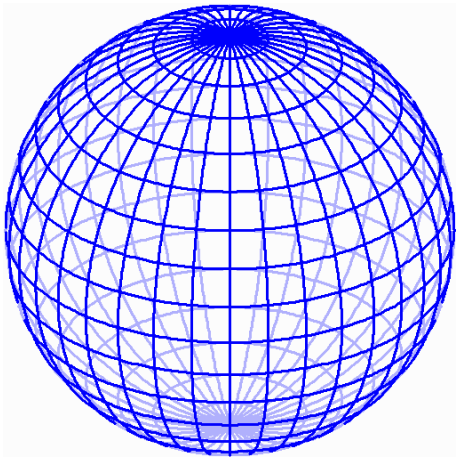
« Sum up in a glance the virgin absence dispersed in this solitude and depart with it, as, to remember a certain place, you pluck one of the magical closed water lilies that suddenly rise up, enveloping nothing in their hollow whiteness, made of untouched dreams, from a happiness that will never be realized, and from the breath that I am now holding for fear of some apparition: silently, rowing away little by little, without breaking the illusion by any sudden shock, and without allowing the rippling visible bubble of foam unfurled by my flight to throw at anyone's arriving feet a transparent likeness of my abducted ideal blossom. »

On the Identity of Spheres

What is the identity of a point *b*

that lives on the surface of a sphere?

Is the identity of *b* trivial or not?



Fundamental Group of the 2-Sphere S^2

- The South Pole b is naturally equipped by the trivial self-identity

$$refl_b : b =_{S^2} b\,.$$

- Now, any other loop p based at b on the surface of S^2 also proves the self-identity of b:

$$p : b =_{S^2} b\,.$$

- Is p homotopic to $refl_b$, i.e. is the proposition

$$p =_{(b=_{S^2}b)} refl_b$$

true?

- Yes! Indeed, any such p can be continuously shrunk to $refl_b$. All loops based at b are homotopic.

- Hence, $[p] = [refl_x]$, which means that the fundamental group $\pi_1(S^2, b)$ is trivial.

The self-identity of a point

on a 2-sphere S^2 is trivial…

But wait… there is some kind of hole in a 2-sphere S^2 !

Fregean vs. Constructive Abstraction

Rather than merging equivalent/homotopic loops

within the same equivalence class

and forget about the homotopies between them

(**Fregean abstraction**)

we can keep homotopic loops distinct

and consider the homotopies between them.

(**Constructive abstraction**)

- In particular, rather than merging p and $refl_b$ within a unique class $[p] = [refl_x]$

we can keep them distinct and consider the homotopies in $p =_{(b=_{S^2}b)} refl_b$.

- In particular, there are two equivalences

$$H_{Right} : p =_{(b=_{S^2}b)} refl_b \qquad\qquad H_{Left} : p =_{(b=_{S^2}b)} refl_b$$

that contract p to the right and to the left respectively.

- Are H_{Right} and H_{Left} equal, i.e. is the proposition

$$H_{Right} =_{(p=_{(b=_{S^2}b)}refl_b)} H_{Left}$$

true?

- No, since the hole within S^2 obstructs the existence of a 3-homotopy between the 2-homotopies H_{Right} and H_{Left}.

Non-Trivial Identity of the Trivial Identity

• The point b has a trivial self-identity uniquely given by

$$refl_b : b =_{S^2} b$$

and any other self-identification p of b is propositionally equal to $refl_b$.

• So, we could conclude that the self-identity of the South Pole b is trivial.

• Now, lets consider the self-identity of the trivial self-identity $refl_b$ of b, which is encoded by the type

$$refl_b =_{(b=_{S^2}b)} refl_b .$$

• As usual, $refl_b$ is endowed with its trivial self-identification

$$refl_{refl_b} : refl_b =_{(b=_{S^2}b)} refl_b .$$

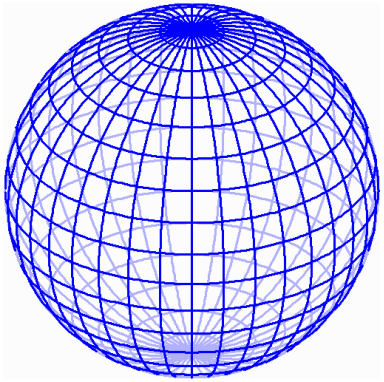
• However, $refl_b$ is also equipped with a non-trivial self-identification given by the surface of the sphere

$$surf : refl_b =_{(b=_{S^2}b)} refl_b$$

such that that the hole within the sphere implies that

$$refl_{refl_b} \neq surf.$$

A 2-sphere S^2



can be described as a point b with a trivial self-identity

$$refl_b : b =_{S^2} b$$

which is identical to itself in a non-trivial manner

$$surf : refl_b =_{(b=_{S^2}b)} refl_b.$$

The Identity of *n*-Spheres

• A 1-sphere or circle S^1 is a point b with a canonical non-trivial identity:

$$l \neq refl_b.$$

• A 2-sphere S^2 is a point b with a trivial identity $refl_b$ with a canonical non-trivial identity:

$$surf \neq refl_{refl_b}$$

• A 3-sphere S^3 is a point b with a trivial identity $refl_b$ with a trivial identity $refl_{refl_b}$ with a canonical non-trivial identity:

$$vol \neq refl_{refl_{refl_b}}$$

Higher Homotopy Groups in General (I)

• Each point in the self-identity type $b =_A b$ (considered itself as a space) is a loop in A based at b.

• We can now consider the fundamental group of the space $b =_A b$ at the point $refl_b$.

• First, the point $refl_b$ is endowed with its trivial self-identity

$$refl_{refl_b} : refl_b =_{(b=_A b)} refl_b$$

• Second, $refl_b$ might be endowed with other self-identifications

$$\gamma : refl_b =_{(b=_A b)} refl_b$$

which are not homotopic to $refl_{refl_b}$.

• In that case, there are « volumetric holes » obstructing the homotopic equalization between the surfaces γ and $refl_{refl_b}$.

Higher Homotopy Groups in General (II)

• The fundamental group of the space $b =_A b$ at $refl_b$ is the group formed by the equivalence classes of self-identifications of $refl_b$.

• This group is the *2nd homotopy group of A at b*, denoted $\pi_2(A, b)$:

$$\pi_2(A, b) \equiv \pi_1(b =_A b, refl_b) \quad \text{(encoding 3-dim. holes)}$$

• By iterating these construction, we can define the higher homotopy groups

$$\pi_3(A, b) \equiv \pi_1(refl_b =_{(b=_A b)} refl_b, refl_{refl_b}) \quad \text{(encoding 4-dim. holes)}$$

$$\pi_3(A, b) \equiv \pi_1(refl_{refl_b} =_{(refl_b =_{(b=_A b)} refl_b)} refl_{refl_b}, refl_{refl_{refl_b}}) \text{ (encoding 5-dim. holes)}$$

Homotopy Levels

« […] nowhere on heaven or on earth

is there anything which does not contain

both being and nothing in itself. »

G.W.F. Hegel, *The Science of Logic*.

« […] the potter who forms sides and bottom on his wheel does not, strictly speaking, make the jug. He only […] shapes the void. For it, in it, and out of it, he forms the clay into the form. From start to finish the potter takes hold of the impalpable void and brings it forth as the container in the shape of a containing vessel. […] The vessel's thingness does not lie at all in the material of which it consists, but in the void that holds. »

M. Heidegger, *The Thing*.

« A construction around emptiness designates the place of the Thing. »

J. Lacan, *The Ethics of Psychoanalysis*.

The « hole affaire »

The existence of a type *X* fills emptiness.

But it does so by exhibiting in a singular manner the emptiness it fills.

So, we can classify types

according to the manner in which they fill and exhibit emptiness.

For instance, a 2-sphere fills emptiness

by exhibiting it in the form of a *hole*,

which is a kind of *localization of emptiness*,

an *absence* that is *there*.

The *n*-spheres define the arquetypal forms of emptiness

(-1)-sphere	Unbounded emptiness
0-sphere	Empty space between two points
1-sphere	Empty space within a circle
2-sphere	Empty space within the sphere

So, we shall classify a type X

by « throwing » n-spherical « lassos » to X.

If one of such « n-lassos » cannot be contracted or filled,

then X exhibits the corresponding form of emptiness.

These non-contractible « n-lassos »

define non-trivial elements in the homotopy group $\pi_n(X)$.

- $\pi_1(X)$ has non-trivial elements when X exhibits emptiness in the form of 1-holes.

- $\pi_2(X)$ is non-trivial when X exhibits emptiness in the form of 2-holes.

- $\pi_n(X)$ is non-trivial when X exhibits emptiness in the form of n-holes.

Is it possible to generalize backwards

and define $\pi_0(X)$ and $\pi_{-1}(X)$?

$\pi_0(X)$ and $\pi_{-1}(X)$

- $\pi_0(X)$ is the number of connected components of X.

$\pi_0(X)$ is non-trivial

when X exhibits emptiness

in the form of external differences

between the multiple components of X.

- $\pi_{-1}(X)$ is trivial when X is non-empty and non-trivial when X is empty.

$\pi_{-1}(X)$ is non-trivial

when X exhibits emptiness

just by not existing at all.

How to fill a n-sphere S^n?

- A sphere S^2 can be filled by a ball D^3.
- A circle S^1 can be filled by a disc D^2.
- Two points S^0 can be filled by a path D^1.
- Emptiness S^{-1} can be filled by a single point $\star$.

0-homotopy group $\pi_0(X)$

- $\pi_0(X)$ is the set of homotopic maps

$$S^0 = (\{-1,1\}, -1) \to (X, x_0)$$

where $$-1 \mapsto x_0$$

which means that the map $S^0 \to (X, x_0)$ is defined by the image of 1.

- Two maps that send 1 to points in the same connected component of X are homotopy equivalent.

- Hence, the set of homotopy classes of maps $S^0 \to (X, x_0)$ is in bijection with the set of connected components of X.

(-1)-homotopy group $\pi_{-1}(X)$

• $\pi_{-1}(X)$ is defined as the homotopy classes of maps

$$S^{-1} = \varnothing \to X\,.$$

• In general $\pi_n(X)$ is trivial if any map

$$S^n \to X$$

can be extended to a map

$$D^{n+1} \to X$$

i.e., if the image of S^n in X can be filled with the corresponding ball D^{n+1}.

• In particular, $\pi_{-1}(X)$ is trivial if

$$S^{-1} = \varnothing \to X$$

can be extended to a map

$$D^0 = \star \to X,$$

that is if X is inhabited.

• Conversely, $\pi_{-1}(X)$ is non-trivial if $\varnothing \to X$ cannot be extended to $\star \to X$, which means that the type X is empty.

n-connectedness

- A pointed space (X, x_0) is said to be n-connected if

$$\pi_i(X, x_0) = 0 \text{ for } i \leqslant n.$$

- The identity of x_0 starts to be non-trivial only after n.

- A space is -1-***connected*** if

$$.\pi_{-1}(X, x_0) = 0 \quad \text{(non-empty)}.$$

- A space is 0-***connected*** if

$$.\pi_{-1}(X, x_0) = 0 \ \text{(non-empty)}$$
$$.\pi_0(X, x_0) = 0 \ \text{(path-connected)}.$$

- A space is 1-***connected*** if

$$.\pi_{-1}(X, x_0) = 0 \text{ (non-empty)},$$
$$.\pi_0(X, x_0) = 0 \text{ (path-connected)}$$
$$.\pi_1(X, x_0) = 0 \text{ (simply-connected)}.$$

Homotopy of Spheres

- n-spheres S^n are $(n-1)$-connected.

- The homotopy groups of spheres $\pi_i(S^n)$ falls in one of the following three cases:

  .when $i < n$, we have $\pi_i(S^n) = 0$.

  .when $i = n$, we have $\pi_i(S^n) = \mathbb{Z}$.

  .when $i > n$, the calculation of $\pi_i(S^n)$ becomes highly non-trivial

  (e.g. Hopf fibration, for which $\pi_3(S^2) \neq 0$).

n-Types

An *n-type* or *n-truncated type*

is a type X such that

$$\pi_i(X, x_0) = 0$$

for all $i > n$

An *n*-type can instantiate forms of emptiness

only in dimensions $i \leqslant n$.

1-Types (aka groupoids)

A 1-type

is a type X such that $\pi_2 = \pi_3 = \ldots = 0$

Such a type

- might be empty or not ($\pi_{-1} \neq 0$),
- it might have different connected components ($\pi_0 \neq 0$).
- each connected component might have 1-holes ($\pi_1 \neq 0$).

0-Types (aka sets)

A 0-type

is a type X such that $\pi_1 = \pi_2 = \ldots = 0$

Such a type

- might be empty or not ($\pi_{-1} \neq 0$),
- it might have different connected components ($\pi_0 \neq 0$),
- but it has no 1-dimensional or higher dimensional holes.

A 0-type is homotopy equivalent to a set of points

(-1)-Types (aka *mere propositions*)

A (-1)-type

is a type X such that $\pi_0 = \pi_1 = \ldots = 0$

Such a type

- might be empty or not ($\pi_{-1} \neq 0$).

- If it is inhabited it has a unique component with no holes of any dimension.

- A mere proposition can only encode a binary or dualistic information (being true or false).

(-2)-Types aka the *One* or the *Universal true*

A (-2)-type

is a type X such that $\pi_{-1} = \pi_0 = \ldots = 0$

Such a type

- cannot not be inhabited ($\pi_{-1} = 0$).

The One is so holeless, that it cannot even not be.

- it has a unique connected component.

- it has no holes of any dimension.

A (-2)-type is a necessarily existing holeless space with a unique component.

It cannot encode any information at all.

The unit type **1** represents the highest level of abstraction

in which all forms of difference are « forgotten »,

which means that

.the ontological difference between being and non-being,

.the ontic differences between multiple beings,

.the extimate self-differences

are simply absent.

Being = One = The Absolute Abstraction

« When abstraction is made from all [differences], [Being] is left over. […] The result of such an abstraction [..] is first of all abstract being, *bein*g in general. […] One can (God be praised!) even abstract from nothing […]. But then, what remains is not nothing, since abstraction would be made even from it and so we would be back at being again. […] Being is the absolute abstraction. »

Pseudo-Hegel, *The Science of Logic*.

Parmenides on the concomitance
between Necessary Being and Universal Truth

« […] that [*it*] *is*, and that [*it*] *cannot not be*,
is the path of Persuasion (for it attends upon truth). »

Parmenides, *Fragment 2*.

« Once in his life Parmenides, probably at a fairly advanced age, had a moment of purest absolutely bloodless abstraction, unclouded by any reality. »

F. Nietzsche, *Philosophy in the Tragic Age of the Greeks*, p. 69.

	Homotopy Groups	Object	Forms of Difference
(-2)-Type	Contractible Space $\pi_{-1} = \pi_0 = \pi_1 = \ldots = 0$	One	No difference at all
(-1)-Type	Contractible if inhabited $\pi_0 = \pi_1 = \ldots = 0$	Mere propositions (Duality)	Ontological difference btw being and non-being
(0)-Type	Simply connected space $\pi_1 = \pi_2 = \ldots = 0$	Sets (Multiplicity)	Ontic differences
(1)-Type	$\pi_2 = \pi_3 = \ldots = 0$	1-groupoid	Self-differences or extimacies
(2)-Type	$\pi_3 = \pi_4 = \ldots = 0$	2-groupoid	Self-self-differences or exextimacies

When we start descending « towards the concrete »

from the summits of the absolute abstraction that posits the One

—i.e. the pure indeterminate Being that is completely disentangled

from any form of (ontological, ontic, or extimate) difference —

identity starts to get mixed with difference

and beings start to exhibit nothingness.

This entanglement between *being* and *nothingness*

produces *becoming*

in the form of entities with non-trivial automorphisms

whose identities can envelop changes,

thereby gaining in plasticity and resilience.

On Becoming

« The truth is neither being nor nothing, but rather that being has passed over into nothing and nothing into being […]. Becoming is this immanent synthesis of being and nothing. »

G.W.F. Hegel, *The Science of Logic*.

The Induction Principle for Equality Types

Definition of Equality in Predicate Logic

- **Reflexivity** $a = a$

- **Principle of substitution *salva veritate***:

  Given a predicate in two variables $P(a, b)$, let's suppose that $a = b$. Then,

$$P(a, a) \Rightarrow P(a, b)$$

From this we can deduce:

- **Symmetry** : $(x = y) \Rightarrow (y = x)$

- **Transitivity**: $(x = y \text{ and } y = z) \Rightarrow x = z$

- **Principle of Indiscernibility of Equals**: equal things have the same properties (we take $P(x, y) \equiv Q(x) \Rightarrow Q(y)$, where $P(x, x)$ always holds. Hence, $x = y$ implies that $Q(x) \Rightarrow Q(y)$).

The Induction Principle for Equality Types in a Nutshell

- Classically: **Principle of substitution *salve veritate*:**

  Given a predicate in two variables $P(a, b)$, let's suppose that $a = b$

  Then,

$$P(a, a) \Rightarrow P(a, b)$$

- In HoTT: **Induction Principle for Equality Types** (aka ***path induction***):

  Given a predicate $P(a, b; p : a =_X b)$, then:

$$P(a, a; refl_a : a =_X a) \Rightarrow P(a, b; p : a =_X b)$$

Towards a constructive *Principle of substitution salva veritate*

• We shall now introduce a constructive or proof-relevant version of the *Principle of substitution salva veritate*:

$$P_a(a) \text{ and } a = b \Rightarrow P_a(b), \text{ where } P_a(x) \text{ is a predicate}$$

• Constructive or proof-relevant upgrading:

$a = b$  $p : a =_X b$

$P_a(x)$  $P_a(x; p)$

$P_a(a)$  $P_a(a; refl_a : a =_A a)$

Base Path Space

- Let's consider the *base path space* type

$$\sum_{x:A} a =_A x$$

encoding the proposition

There exists a term x such that x is equal (i.e. path-connected) to a.

- A proof/term of this proposition/type is a pair

$$(x, p : a =_A x).$$

- This type is the *space of all paths based at a*.

- This type is contractible with center of contraction $(a, refl_a)$ and every contractible type is equivalent to the type $\mathbf{1}$.

- The induction principle for $\mathbf{1}$ says that any predicate $P_a(x, p : a =_A x)$ on the type $\sum_{x:A} a =_A x$ holds for every $(x, p : a =_A x)$ if it holds for the center of contraction $(a, refl_a)$.

Consequences of the induction principle for equality

- Equality is ***symmetric***:

$$(x = y) \Rightarrow (y = x)$$

Equalizations/paths can be inverted

- Equality is ***transitive*** (concatenation of paths):

$$(x = y \text{ and } y = z) \Rightarrow (x = z)$$

Equalizations/paths can be concatenated

Principle of the Indiscernibility of Equals

If we have a predicate $Q(x)$ over A and a proof

$$p : a =_A x,$$

then we have a function

$$p_* : Q(a) \to Q(x)$$

that sends proofs of $Q(a)$ to proofs of $Q(x)$ for any predicate Q,

which means that $Q(x)$ is also true.

Terms that are propositionally equal are indiscernible.

Equals share properties

There is no equality without common property.

Geometric Interpretation of the Indiscernibility of Equals as a Path-Lifting

Let's suppose that

$$p_* : Q(a) \to Q(x)$$

is such that

$$u : Q(a) \mapsto p_*(u) : Q(x)$$

The path $p : a =_A x$ can be *lifted*

to a path in the whole space of the fibration $Q(x)$ over A

starting at $u : Q(a)$ and ending at $p_*(u) : Q(x)$.

Equalizations/paths can be used as tracks to transport properties.

Type Theory	Homotopy Theory	Category Theory
Terms	Points	Objects
Equalizations	Paths	1-isomorphisms
$refl_a$	Constant Path γ_a	id_a
Higher Equalizations	Homotopies	Higher isomorphisms
Symmetry of equality	Reversibility of paths	Invertibility of morphisms
Transitivity of equality	Concatenation of paths	Composition of morphisms
Types	Homotopy Types	∞-groupoids

Since all the category-theoretical 1-morphisms defined by the type-theoretical equalizations are reversible, we could guess that the category induced by a type is a 1-groupoid.

- However…

.the associativity of compositions:

$$(p \circ q) \circ r = p \circ (q \circ r)$$

.the fact that trivial self-identities should act as neutral elements for composition:

$$id_x \circ p = p \qquad\qquad p \circ id_y = p$$

.and the property that defines isomorphisms:

$$p^{-1} \circ p = id_y \qquad\qquad p \circ p^{-1} = id_x$$

do not hold as strict equalities !!!

The groupoid laws hold as propositional equalities (i.e. through coherators)

- **Associators**: given $p : x =_A y$, $q : y =_A z$ and $r : z =_A w$, we have

$$associator_{(p,q,r)} : (p \circ q) \circ r =_{(x=_A w)} p \circ (q \circ r) .$$

- **Unitors**: given $p : x =_A y$, we have

$$left - unitor : refl_x \circ p =_{(x=_A y)} p$$

and

$$right - unitor : p \circ refl_y =_{(x=_A y)} p$$

- **Inverters**: given $p : x =_A y$, we have

$$left - inverter : p^{-1} \circ p =_{(y=_A y)} refl_y$$

and

$$right - inverter : p \circ p^{-1} =_{(x=_A x)} refl_x$$

Univalence Axiom

Homotopy Hypothesis

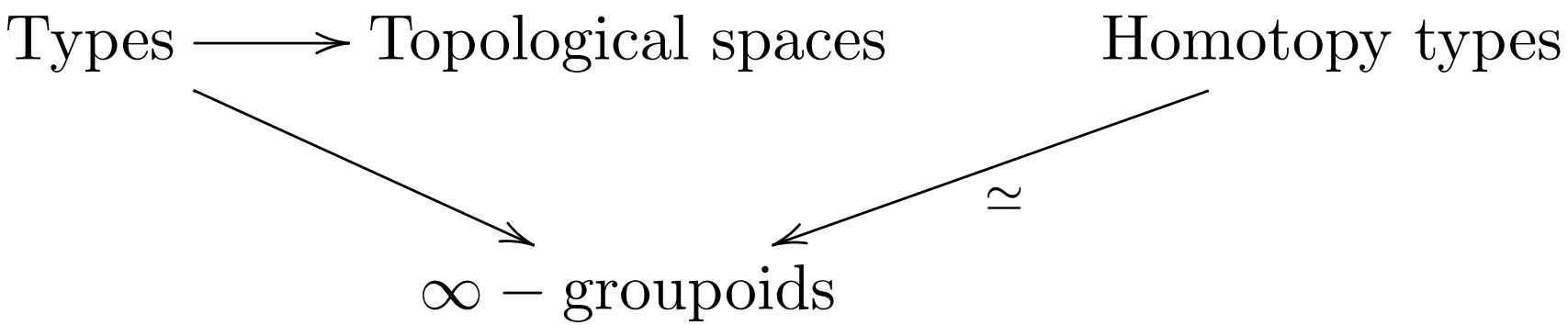
- Given a topological space X, we can define its *fundamental* ∞*-groupoid* $\pi_\infty(X)$.

- Reciprocally, an ∞*-groupoid* defines a space up to homotopy equivalence through a construction called *geometric realization*.

- Topological spaces up to homotopy equivalence are called *homotopy types*.

- So, we have an equivalence (called « homotopy hypothesis »):

Homotopy types  **∞-groupoids**


Types
Topological spaces
Homotopy types
$\simeq$
$\infty - \text{groupoids}$

In order to close the triangle between

.types,
.geometric objects,
.∞-groupoids,

Voevodsky axiomatically introduced a criterion of identity for types

called ***univalence,***

which amounts to state that two types are propositionally equal

if they are homotopy equivalent as topological spaces.

Types $\leftrightarrow$ Homotopy Types

Homotopy (Type Theory) $\leftrightarrow$ (Homotopy Type) Theory

Univalence Axiom

- Types are terms of a type of types call *universe* $\mathcal{U}$.

- Given two types $A, B : \mathcal{U}$ we can form the equality type

$$A =_{\mathcal{U}} B\,.$$

- A proof $p : A =_{\mathcal{U}} B$ is a path in the space $\mathcal{U}$ between A and B.

- The Univalence Axiom states that types that are *homotopy equivalent* (when considered as topological spaces) are *propositionally equal*.

- Symbolically,

$$(A \simeq B) \to (A =_{\mathcal{U}} B)$$

or

$$(\ \simeq\ ) \to (\ =_{\mathcal{U}}\ )$$

Homotopy types as *abstract shapes*

• Homotopy types are ***shapes*** since they result from an abstraction process acting on topological spaces.

• Homotopy types ***abstract*** objects because they result from these topological spaces by forgetting all properties that are not homotopy equivalent.

• A homotopist is a persons that does not make the difference between

.a *punctured plane*,
.a *cylinder*,
.a *Mobius band*,
.a *circle* ,

which are all presentations of the same homotopy type, which is the abstract shape common to all these spaces.

Homotopy Type Theory Ontology

The fundamental mathematical objects are ***abstract shapes***.

Space as a *principium individuationis*

According to HoTT,

the fundamental mathematical entities

are not *sets*

but *abstract geometric shapes*.

Why does the notion of space

play such a fundamental role in mathematics?

• A set X is a homotopy type such that

$$\pi_1(X) = \pi_2(X) = \ldots = 0$$

where the only nontrivial homotopy by be in $\pi_0(X)$ (encoding the different connected components of X).

Set-theoretic foundations are 0-truncated homotopic foundations

• Homotopic foundations stem from set-theoretic foundations when we relax the 0 -truncation condition

$$\pi_i(X) = 0 \text{ for all } i > 0.$$

• This amounts to blow up each grain of the set-theoretic dust into a connected space that might be topologically non-trivial at every dimension i>0.

• According to HoTT,

Mathematics is not built on a ***set-theoretic dust***,

but rather on a kind of ***topological foam***.

- A topological space combines two dimensions:

  .**Discreteness**:

  .Dimension of *differentiation* or *separation* (related to its different connected component).

  .**Continuity**:

  .Dimension of *identification* or *repetition* (related to the existence of different points that can be path-connected).

- Set-theoretic foundations

  .truncates the continuous identifications between indiscernibles,
  .and only retains the discrete separation.

- The introduction of the notion of space into the very foundations of mathematics makes possible the existence of indiscernible terms that are numerically different,

… that is, it liberates mathematics from the grips of the Principle of the Identity of Indiscernibles.

« Since the mere Here is nothing by itself that might differ from any other Here, space is the *principium individuationis*. It makes the existence of numerically different things possible which are equal in every respect. […] Leibniz infers from this the ideality of space and time; for they violate the principle of the identity of indiscernibles, which […] he postulates as necessary in the domain of substances […] »

H. Weyl, *Philosophy of Mathematics and Natural Science*, 1949.

On Abstraction

Very briefly, to abstract means

to ***bracket*** or ***« forget » certain differences***

that can be considered irrelevant in a given context.

By « forgetting » certain differences,

we ***equalize objects that are strictly speaking different***.

On the one hand, bracketing differences allows us

- to recognize patterns, structures, regularities, forms;
- to engage in different forms of exchange processes (linguistic, economic, etc.);
- to reidentify entities through change;
- to enlarge our toolkit of concepts with new *abstracta*.

On the other hand, bracketing differences entails different forms of violence exerted against all what is abstracted away, like

- culturally specific contexts (in the constitution of *universality*),
- subject-dependent descriptions (in the constitution of *objectivity*),
- singular features of individuals (in *taxonomies* and *statistical analysis*),
- concrete dimensions of human labor and its products (in *economic exchange processes*).

In short, the main risk is that of

hypostatizing abstract concepts

by forgetting that they are the result

of a mental or real process of abstraction.

If abstract concepts result from an act by means of which

we methodically « forget » certain differences,

the risk is that of ***forgetting that we have forgotten these differences.***

The Critique of Abstract Reason

- One-sided *either-or* nature of the abstract concepts of the understanding (G.W.F. Hegel).

- Reification and fetishization (K. Marx, G. Lukács).

- « Vicious abstractionism » (W. James).

- « Fallacy of misplaced concreteness » (A.N. Whitehead).

- the conversion of *types* into *stereotypes* (M. Jackson).

- the conflation of the *map* and the *territory* (A. Korzybski).

Fregean Abstraction

• Mathematically, abstraction is modeled through the so-called *Fregean abstraction*, which proceeds by

.enriching a set X of « concrete » individuals with an equivalence relation $\sim$.

.projecting X to a set of abstracta $X/\sim$:

$$q : X \to X/\sim$$

$$x \mapsto [x]$$

A Critique of Fregean Abstraction

- Fregean abstraction relies on the notion of *equivalence relation*, which

  .can only encode the mere existence of an equivalence between two terms,

  .but not the concrete (possibly multiple) equivalences between them.

- All in all, Fregean abstraction necessarily truncates the structure of (higher) equivalences.

- This truncation might entail different forms of pathologies.

Fregean abstraction falls under

what W. James called « vicious abstractionism ».

« Vicious abstractionism » is a way of using concepts

that ***simplifies*** the domain of « concrete » individuals on which it acts

and hypostatizes the corresponding abstracta.

Fregean abstraction is a simplification of the original domain

since it merges different individuals within the same equivalence class

and « forgets » the equivalences between them.

« [The original function of concepts is] to enlarge mentally our momentary experiences by adding to them the consequences conceived; but unfortunately, that function is not only too often forgotten […] but is often converted into its exact opposite, and made a means of diminishing the original experience by denying (implicitly or explicitly) all its features save the one specially abstracted to conceive it by. […] Let me give the name of 'vicious abstractionism' to a way of using concepts which may be thus described: We conceive a concrete situation by singling out some salient or important feature in it, and classing it under that; then, instead of adding to its previous characters all the positive consequences which the new way of conceiving it may bring, we proceed to use our concept privatively; reducing the originally rich phenomenon to the naked suggestions of that name abstractly taken, treating it as a case of 'nothing but' that concept, and acting as if all the other characters from out of which the concept is abstracted were expunged. Abstraction, functioning in this way, becomes a means of arrest far more than a means of advance in thought. It mutilates things […] *The viciously privative employment of abstract characters and class names* is, I am persuaded, one of the great original sins of the rationalistic mind. »

W. James, *The Meaning of Truth*.

« The falsehood of an unleashed rationality running away from itself, the recoil of enlightenment into mythology, is rationally definable. [...] The *ratio* becomes irrational [...] where it runs counter to the meaning of thought by hypostasizing its products, the abstractions. »

T.W. Adorno, *Negative Dialectics*, p. 34.

Hegel against « cancel culture »

« Who thinks abstractly? The uneducated, not the educated. I have only to adduce examples for my proposition: A murderer is led to the place of execution. For the [uneducated] he is nothing but a murderer. This is abstract thinking: to see nothing in the murderer except the abstract fact that he is a murderer, and to cancel all other human existence in him with this simple quality. »

G.W.F Hegel, *Who Thinks Abstractly?* (1807).

Is it possible to understand abstraction as

an ***extension***,

an ***enrichment***,

an ***expansion***,

of human experience,

rather than a ***simplification*** of it?

Towards a Constructive Abstraction Principle

« To build up an abstract concept from a raw flow of data, one must disregard inessential details; in other words, to simplify the complexity of concrete reality one must idealize over it, and this is obtained by ‘forgetting’ some information. To forget information is the same as to destroy something, in particular if there is no possibility of restoring that information [...] So to abstract involves a certain amount of destruction; our principle is that an abstraction is constructive, that is a reliable tool in getting knowledge which is faithful to reality, not when information is kept as much as possible, but when it is ‘forgotten’ in such a way that it can be restored at will at any moment. This after all is the test to show that an abstraction does not lead astray from reality; that is, that it preserves truth. It is clear that the first step, and often the only one, to be able to restore what has been ‘forgotten’ is to know, to be aware of, what has been forgotten, and to keep control of it. So the second principle is that constructivism does not consist of an a priori self-limitation to full information, which would tie constructivism with reluctance to abstraction (as was the case around the 1920s when finitism and intuitionism were somehow identified), but rather in the awareness of the destruction which has been operated to build up a certain abstract concept. »

G. Sambin and S. Valentini, *Building up a toolbox for Martin-Lof's type theory: subset theory*, 1998, p. 224.

We want to introduce a conception of abstraction

that does not truncate

the (higher) structure of equivalences.

To do so, the first step is not to loose

the distinction between equivalent terms.

If we keep equivalent terms separated

but we « forget » their differences,

we shall obtain terms

that are numerically different

and qualitatively identical.

These terms violate

Leibniz’s Principle of the Identity of Indiscernibles.

Abstraction According to Leibniz

- Leibniz himself maintained that abstract notions violate the Principle of the Identity of Indiscernibles.

- According to Leibniz, abstraction produces indiscernible objects:

« [...] perfect similarity is found only in incomplete and abstract notions, where things are considered only in a certain respect, but not in every way, as, for example, when we consider shapes alone, and neglect the matter that has shape. And so we can consider two similar triangles in geometry, even though two perfectly similar material triangles are nowhere found. »

G.W. Leibniz, *Primary Truths.*

« The decision to abstract from […] generates indistinguishability in the initial universe of objects. »

I. Angelelli, Frege and Abstraction, *Philosophia Naturalis*, 21 (1984).

Now, the type-theoretic notion of *propositional equality* $=_A$

precisely relates

numerically different objects

in such a way that it forces their indiscernibility

(since propositional equalities are like « tracks »

along which properties can be transported)

Constructive Principle of Abstraction

Rather than merging equivalent individuals within the same equivalence class,

we shall define a type X such that equivalent individuals a and b

become propositionally equal terms

$$a =_X b$$

and equivalences

$$a \xrightarrow{\simeq} b$$

become proofs of their equality

$$p : a =_X b$$

Constructive abstraction converts a partial similarity encoded in an equivalence

into an absolute indiscernibility encoded in a propositional equality.

Constructive Abstraction in a Nutshell

Rather than *removing differences* (simplification),

we *add equalizations* (enrichment).

Fregean vs. Constructive Abstraction

- ***Fregean abstraction***:

  Abstraction is ***a simplification***

  of concrete experience.

- ***Constructive abstraction***:

  Abstraction is ***an additional structure of equalizations***

  understood as an ***enrichement*** of concrete experience.

Fregean Abstraction Principle

(equivalences-to-strict-equalities principle)

$$a \sim b \iff [a] = [b]$$

Constructive Abstraction Principle:

(∞-equivalences-to-propositional-equalities principle)

$$a \sim_\infty b \iff \hat{a} =_X \hat{b}$$

Making « abuses of notation » rigorous

Given any form of equivalence $a \simeq b$

(like for instance $2 \times 2 \simeq 3 + 1$),

we can transform the equivalences

into propositional equalities btw a and b understood as terms of a type X.

By doing so, the abuse of notation becomes rigorously true

as a propositional equality

$$a =_X b$$

The Univalence Axiom *qua* Example of Constructive Abstraction

Two topological spaces A and B that are homotopy equivalent

$$A \simeq B$$

become propositionally identical

$$A =_{\mathcal{U}} B$$

as terms of the universe $\mathcal{U}$.

Cell Complexes

and

Homotopic Amalgamations

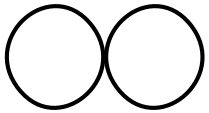
On Bricolage

• We shall consider a very concrete form of abstraction, namely *bricolage* or *handiwork*.

• A fundamental operation in bricolage is *gluing* or *amalgamating* two objects.

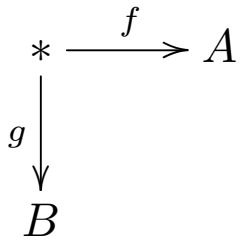
• For instance, gluing two circles S^1_1 and S^1_2 along two points $a \in S^1_1$ and $b \in S^1_2$ can be understood as an operation of abstraction that « forgets » the difference between a and b.

• In mathematical terms, we are forcing the points a and b to become strictly equal.

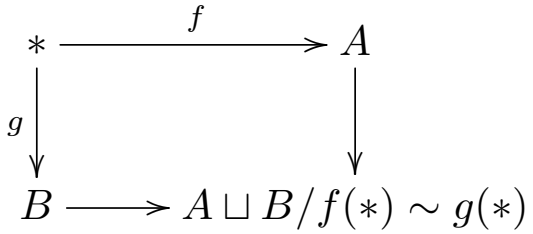
• By doing so, we obtain an *eight* 

Amalgamated sums

• In order to *amalgamate* A and B by gluing a point a of A with a point b of B, we select these points by means of two maps



• The *amalgamated sum* is defined by the diagram



Amalgamating templates

• More generally, we can amalgamate A and B along figures X (called *amalgamating template*) other than points.

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & A \\ {\scriptstyle g}\downarrow & & \downarrow \\ B & \longrightarrow & A \sqcup B / f(x) \sim g(x) \end{array}$$

• For instance, we could glue two disks D^2 along their circular boundaries S^1, therebt-y obtaining a 2-sphere S^2.

$$\begin{array}{ccc} S^1 & \xrightarrow{\quad f \quad} & D^2 \\ {\scriptstyle g}\downarrow & & \downarrow \\ D^2 & \longrightarrow & S^2 \end{array}$$

Cell complexes

• A particular kind of spaces, called *cell complexes,* can be constructed by successively gluing n-balls D^n through their borders S^{n-1}.

$$\begin{array}{ccc} S^{n-1} & \longrightarrow & D^n \\ \downarrow & & \\ X & & \end{array}$$

• We start with a set X_0 of points, the 0-skeleton. Then,

.We add 1-balls D^1 by identifying the border of each disk, namely a 0 -sphere S^0 with points in X_0, thereby obtaining the 1-skeleton X_1.

.We add 2-balls D^2 by identifying the border of each disk, namely a 1 -sphere S^1 with points in X_1, thereby obtaining the 2-skeleton X_2.

• The n-skeleton X_n is obtained from the $(n-1)$-skeleton by adding n-balls to X_{n-1}.

Examples of cell complexes

- 1-**sphere** S^1 (circle):

  .one 0-ball $\star$,
  .one 1-ball D^1, where its boundary S^0 (its two endpoints) is glued to $\star$.

- 2-**sphere** S^2:

  .one 0-ball $\star$,
  .one 2-ball D^2, where the boundary S^1 of D^2 is glued to $\star$.

- **Torus**:

  .one 0-ball $\star$,
  .two 1-balls D^1, where their boundaries S^0 are glued to $\star$ (thereby obtaining two circles glued by a point).
  .one 2-ball D^2 (visualized as a square), where each pair of opposite sides is identified to one of the circles.

- **Möbius Band**:

  .two 0-balls $\star$,
  .three 1-balls D^1.
  .one 2-ball D^2 (visualized as a rectangle).

This construction of geometric objects

by amalgamating « elementary » building blocks

relies on Fregean abstraction,

since the difference between the attaching points

is simply removed.

Let’s suppose now that we want to glue abstract shapes (homotopy types).

Since we know how to glue spaces,

we can glue abstract shapes by gluing their presentations through spaces.

To do so, we have to be sure that the gluing operation

does not depend on the chosen presentation of the abstract shapes.

Amalgamated Sums Do Not Respect Homotopy Equivalence

On the one hand,

$$\begin{array}{ccc} S^1 & \longrightarrow & D^2 \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^2 \end{array}$$

Whereas on the other

$$\begin{array}{ccc} S^1 & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & * \end{array}$$

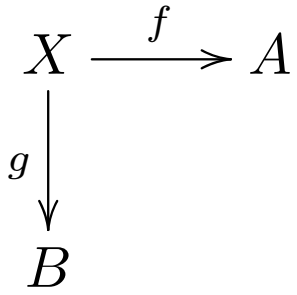
Even if D^2 is homotopy equivalent to $*$.

Homotopic Bricolage

- **Constructive abstraction**:

Rather than removing the unwanted differences, we add equalizations.

- Let's suppose that we want to perform the following amalgamation:



- Rather than establishing a strict equality between $f(x) \in A$ and $g(x) \in B$ for each $x \in X$…

… we shall add one path I_x between $f(x) \in A$ and $g(x) \in B$ for each $x \in X$.

Example revisited

Now, the homotopic amalgamated sums

$$\begin{array}{ccc} S^1 & \longrightarrow & D^2 \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^2 \end{array}$$

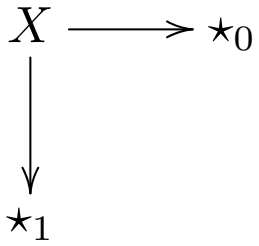
and

$$\begin{array}{ccc} S^1 & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^2 \end{array}$$

yield the same result, namely the 2-sphere S^2

Suspensions

• Let's consider the amalgamation of two points $\star_0$ and $\star_1$ along X:



• The Fregean amalgamating sum yields simply a point.

• The homotopic amalgamating sum adds an interval I_x uniting $\star_0$ and $\star_1$ for each $x \in X$.

• The resulting object is called the *suspension of* X and denoted ΣX.

Examples of Suspensions

- The suspension of S^0 is a circle S^1:

$$\begin{array}{ccc} S^0 & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^1 \end{array}$$

- The suspension of a circle S^1 is a sphere S^2:

$$\begin{array}{ccc} S^1 & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^2 \end{array}$$

- In general, the suspension of a n-sphere S^n is the $n+1$-sphere S^{n+1}:

$$\Sigma S^n = S^{n+1}$$

$$\begin{array}{ccc} S^n & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^{n+1} \end{array}$$

- Generalizing backwards, S^0 can be obtained as a suspension of $\varnothing$. Hence, $S^{-1} \equiv \varnothing$

$$\begin{array}{ccc} S^{-1} \equiv \emptyset & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & S^0 \end{array}$$

Emptiness is spherical



« In the hollow a kangaroo can hide without forfeit of its bounding joy. »

José Lezama Lima, *Pavilion of Nothingness*.

On Go

Life requires *plasticity*, *resilience*,

which in turns requires embodying *non-trivial forms of identity*,

which in turns requires *exhibiting different forms of emptiness*,

including those associated to the possibility of *enveloping extimate holes*.